Quality Assurance Project Plan for Stormwater Characterization Monitoring

Conducted Under Section S8.D of the

Phase I Municipal Stormwater Permit

by

Clark County

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Clark County Department of Environmental Services

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Title and Approval Sheet

Quality Assurance Project Plan

Phase I Municipal Stormwater NPDES Permit

Internal Review and Approval

External Review and Approval

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Quality Assurance Project Plan

This Quality Assurance Project Plan (Plan) was prepared by U.S. Geological Survey and revised by Clark County's Department of Environmental Services Clean Water Program. The Plan describes the quality assurance and quality control (QA/QC) procedures for field activities and laboratory analyses associated with stormwater characterization monitoring conducted by Clark County, as required by Section S8.D of the National Pollutant Discharge and Elimination System and State Waste Discharge General Permit for Discharges from Large and Medium Municipal Separate Storm Sewer Systems (Phase I Permit or Permit). The primary goal of this Plan is to assure the quality and integrity of the collected samples, the representativeness of the results, the precision and accuracy of the analyses, the completeness of the data, and the delivery of defensible products and decisions.

This document was developed with guidance from the Department of Ecology; specifically, Guidelines for Preparing Quality Assurance Project Plans for Environmental Studies (Ecology 2004), and Quality Assurance Project Plan Guidance and Template For Conducting Stormwater Monitoring To Meet Requirements of S8.D of the Phase 1 Municipal Stormwater Permit (August 2008).

This Plan is organized into the following sections:

A. Goals and objectives of stormwater monitoring program
B. Type, quality, and quantity of data needed to meet program objectives
C. Sampling and measurement procedures needed to acquire data needed
D. Quality assurance and quality control procedures to ensure the Plan is implemented as prescribed
E. Assessment procedures to determine if the data conform to the specified criteria and will satisfy the program objectives, and the analysis and format for presentation of the results

Large tables that are for frequent reference during the life of the monitoring program are located in a Tables section following the text, noted with a "T" prefix.

A series of Standard Operating Procedures (SOPs) are being developed by collaborating Phase I Permittees, Clark County, and consultants to provide guidance to users of this Plan. These are described in Table T 1.

Section A. Goals and Objectives of the Monitoring Program

This section covers basic program management, including history and objectives, delegation of responsibilities, and other details to ensure that the program is well defined and understood by all participants. The following elements are included:

1. Background
2. Program description
3. Organization and schedule

1. Background

Stormwater monitoring is required to be conducted by Clark County by Section S8.D of the Phase I Permit, as modified by the Final Settlement Agreement between the Washington Department of Ecology and Phase I Permittees. To address this monitoring requirement, Stormwater Monitoring will be conducted at three locations in Clark County and will include collection of event-based and grab samples for chemical analysis, seasonal first-flush toxicity testing, and an annual sediment sample for chemical analysis.

1.1. Permit overview and monitoring requirements

The Washington State Department of Ecology (Ecology) issued the current Phase I Permit on January 17, 2007 with modifications on June 17, 2009. The Phase I Permit applies to all entities in Washington State that are required to have stormwater permit coverage under current (Phase I) U.S. Environmental Protection Agency (EPA) stormwater regulations. This includes unincorporated portions of counties whose populations exceed 100,000 in the 1990 census, specifically Clark County.

In accordance with the Phase I Permit, each permittee is required to develop and implement a comprehensive long-term monitoring program consistent with Special Condition S8 of the Permit. In general, the required monitoring program shall include the following components:

- Stormwater characterization monitoring (S8.D)
- Targeted stormwater management program effectiveness monitoring (S8.E)
- Stormwater treatment and hydrologic management best management practices evaluation monitoring (S8.F).

A separate Plan is developed for each component of the monitoring program; this Plan provides details on the stormwater characterization monitoring component only.

Ecology requires each permittee to conduct a comprehensive long-term stormwater characterization monitoring program in order to provide a baseline for future reduction of stormwater pollutants discharged to receiving waters. The information obtained from this program will provide a critical starting point for evaluating the effectiveness of stormwater management practices, as well as increasing understanding about potential seasonal trends in stormwater character.

1.2. Stormwater monitoring

In accordance with Section S8.D of the Permit, Clark County is required to monitor three stormwater conveyances, representing three separate land-use categories as follows:

- Commercial
- High-density residential
- Low-density residential

Ideally, the area served by each conveyance selected to represent a specific land use should consist of at least 80% by area of that specified land use (per Permit section S8.D.1.a). The primary objective of this stormwater monitoring program is to characterize these representative sites so that long-term trends can be observed.

Stormwater at the selected sites will be monitored as follows, according to the Permit requirements in S8.D:

- Continuous flow monitoring for the first year, and during all sampled storm events thereafter
- Flow-weighted composite and grab samples from baseflow (as appropriate) and each qualifying storm event for chemical analyses
- Samples for toxicity testing of seasonal first-flush storm events
- Annual cumulative composite sediment samples for chemical analysis

The type, number, and criteria used to select storm events for sampling and the parameters for analysis are consistent with Section S8.D of the Permit, and are discussed in further detail in the following sections of the Plan.

1.3. Monitoring challenges

Characterizing municipal stormwater runoff is not trivial because of several features of stormwater that distinguish it from water transported in streams or stored in lakes. Most importantly, stormwater potentially contains a wide variety of chemical pollutants whose concentrations depend upon land use and other regional and/or local conditions. Different types of storm events throughout the year cause further variability in contaminant concentrations. Moreover, sampling to characterize stormwater is complicated by the unpredictability of storm events and the seasonal and sporadic nature of surface runoff. Additional logistical challenges occur in sampling stormwater where some sites must be located within an urban setting.

1.4. Study area

The study area includes urban and rural land in unincorporated Clark County (Figure 1). Much of Clark County is within the northern-most portion of the Willamette Valley Ecoregion. Land use can be described as suburban within the developed areas and rural residential outside of the developed parts of the Urban Growth Area. Urbanization during the late 20th Century and early 21st Century has converted much of the farmland near Vancouver into residential subdivisions and small commercial areas along existing arterials and highways. Generally, the study area is comprised of gently rolling hills, about 200 to 300 feet above sea level, which are cut by small stream systems draining to Salmon Creek or Lake River. Late Ice Age Cataclysmic Flood Deposits underlay the study area and provide a source for fine-grained sediment.

Commercial site The commercial site represents a segment of Highway 99 and a group of older highway-oriented businesses such as auto repair shops and apartment buildings (Figure 2). The basin is located east of I-5, and drains an area of about 25 acres. The outfall is located in a pipe that drains directly to a piped section of Cougar Creek. Monitoring access, just upstream from the outfall, is provided via an existing manhole on

the sidewalk along the east side of Highway 99 just south of 82nd Street. Confined-space certification is needed for monitoring work below the sidewalk.

High-density residential site The outfall to measure the high-density residential stormwater site drains multiple neighborhoods, with a drainage area of approximately 240 acres, and is located near the boundary between Felida and Lakeshore Neighborhoods north of Vancouver (Figure 3). The area is characterized by mixed neighborhoods with an average lot size less than ¼ acre, and represents high density residential areas of Clark County in general. The outfall is a 36-inch metal pipe that drains directly into Vancouver Lake. It is suitable for monitoring and is located in an accessible manhole near 11100 NW 36th Avenue. Confined-space certification will be needed to access this outfall.

Low-density residential site The low-density residential site drains about 45 acres of rural residential homes sited on lots that are five acres in size (Figure 4). Other than buildings and road surfaces, the catchment is mostly covered with ungrazed, grassy pasture or hay fields. The catchment has a relatively steep topography and drains to a set of culverts, which outfall into a ditch forming the head waters to a tributary of Packard Creek. Monitoring will occur at the very upper end of this tributary stream as it transitions from the outfall ditch west of the intersection of NW 11th Avenue and NW 184th Street. This outfall site does not require confined space certification.

1.5. Parameters of concern

Local land use conditions are important factors determining the likely importance of specific pollutants in stormwater runoff. In addition to activities related to land use, other pollution sources that may add contaminants to stormwater include vehicular traffic conditions, atmospheric deposition, and illicit discharges and spills. The parameters of concern for this program are defined by Ecology in the phase I permit, and include conventional pollutants, nutrients, metals, and organic compounds measured in both stormwater and sediment (Table T 2). All constituents have a history of association with stormwater discharge from urban environments, and pose some threat to ecological conditions in receiving waters either through disruption of habitat or introduction of toxic conditions.

2. Program description

This section describes the goals and objectives of the program in detail, including the information requirements, specific target characteristics to be monitored, as well as the data required to meet the program objectives. Study area boundaries and practical constraints that will need to be considered are also briefly discussed.

2.1. Goal and objectives

The goal of this program is to collect and characterize stormwater and sediment samples from representative conveyances within Clark County to fulfill the requirements of Section S8.D of the Phase I Permit. These requirements are designed to acquire knowledge of pollutant loadings and changes in conditions over time for representative land uses in order to facilitate generalizations across the range of Permittees' jurisdictions. Particular emphasis is given to developing the capacity to evaluate whether stormwater management programs are achieving the goal of reducing pollutant loads and thereby protecting water quality in receiving waters.

The specific objectives of the program include:

1. Identify three stormwater basins and outfalls to sample within Clark County that are representative of the specified land uses covered by the Phase I Permit (S8.D.1)
2. Collect representative composite and grab samples during baseflow, as applicable, and stormflow events from each selected outfall (S8.D.2.a-c,e,g)
3. Collect a representative annual sediment sample from each selected outfall (S8.D.2.f)
4. Collect a dry-season first-flush stormwater sample for toxicity testing (S8.D.2.d)
5. Validate and report the sampling results to Ecology (S8.H.1.a)

2.2. Information requirements

Information needed to meet the study objectives includes:

- Land use and drainage area of each selected drainage basin
- Concentrations of constituents of concern in samples from each outfall
- Continuous record of rainfall data, especially prior to and during sampled storm events, including antecedent dry period and total rainfall during each event
- Continuous record of outfall flow data (storm and base flow, where appropriate), especially flow data during sampled storm events

- Toxicological results and associated analytical constituent concentrations

2.3. Data collection

The sampling design for stormwater monitoring under Section S8.D of the Permit contains three primary components that will be conducted at the selected monitoring sites through the remainder of the Phase I Permit cycle:

- Stormwater characterization
- Toxicity sampling
- Sediment sampling

Stormwater sampling Flow-weighted composite and grab stormwater samples will be collected from each site during eleven to fourteen qualifying storm events (defined in Qualifying Storm Events section) during each water year (S8.D.2.a). Grab samples will be collected as early as possible in the storm during each targeted storm event (S8.D.2.e).

Continuous flow data will be collected at each site for one year to define rainfall-runoff relationships at each outfall (S8.D.2.b). Additionally, continuous flow data will be collected during all sampled storm events for the duration of the program. Total annual and wet/dry seasonal pollutant loads will be calculated for each required constituent at each sampling site based on event mean concentrations, including contributions from baseflow as appropriate (S8.H.1.a).

Toxicity sampling Samples for toxicity testing will be collected with composite sampling equipment at each monitoring site during the late summer or early fall in order to sample a first-flush storm event (S8.D.2.d). Stormwater will be collected concurrently for chemical analysis in order to identify toxic substances that may be present, as well as to identify substances that may interfere with the toxicity test.

Sediment sampling In-line sediment traps will be utilized to collect annual sediment samples at each monitoring site (S8.D.2.f). Sub-samples will be collected as appropriate throughout the year and composited together at the end into a single sample from each site for analysis.

2.4. Target population

Sampling is designed to describe target populations, in this case, stormwater characteristics for each category of land use required by the Permit. These include:

- Rate and volume of stormwater and baseflow
- Specific constituents concentrations and loads in stormwater and concentrations in sediments carried by stormwater (Table T 2)

2.5. Study area boundary

The study area boundary describes the unincorporated areas of Clark County covered under the jurisdiction of the Phase I Permit. Monitoring sites were selected for basins according to the land-use requirements specified in the Permit: "each site must represent a discernible type of land use, but not a single industrial or commercial complex. Ideally, to represent a particular land use, no less than 80% of the area served by the outfall or conveyance will be classified as having that land use." (S8.D.1) Specific details on the criteria used to select the sites are presented in Section 7.

2.6. Practical constraints

For some sites, monitoring may be constrained by specific characteristics of the storm drain system and hydraulics. The choice of flow measurement methods and the accuracy of flow measurement are limited by pipe diameter and slope, water depth, and backwater conditions. For those sites where the most suitable location for monitoring is a manhole structure, confined space entry will be required to install, operate, and maintain monitoring equipment. Features of the storm drain will also constrain the deployment of sediment traps, especially where drain pipes have small diameters. Other activities at the sites will similarly constrain monitoring by field crews, especially if sampling sites are located in or near active parking areas where steps would need to be taken to ensure continual access.

The ability of automatic samplers to collect representative samples of stormwater will be constrained largely by the storage capacity of the sampler, which determines the frequency and duration of sample collection. Similarly, deployment of sediment traps will be complicated by the difficulty in predicting how much sediment will be captured at a particular site. If insufficient sample is collected, it may not be possible to conduct all

the desired physical and chemical analyses. Alternatively, if sediment yields are higher than expected during specific storm events, the trap may overfill and some sediment will not be captured.

The limited reliability of precipitation forecasts also creates inherent logistical challenges for stormwater monitoring. Mobilization of field crews for a potential sampling event will certainly be restricted to no more than a few days prior to the storm, often only a few hours.

Finally, some chemical constituents cannot be collected using automatic sampler and must be collected manually. Because these samples are required to be collected early in the storm, field crews must be mobilized on short notice. Scheduling will be difficult, especially for storms that begin at night or during weekends or holidays. Collecting these grab samples will be subject to staff availability.

2.7. Decision making

The results of this monitoring program will not be utilized to make specific decisions. Nonetheless, these results will allow Ecology to assess the progress that is being made by stormwater management programs in reducing the transport of pollutants via stormwater, thereby achieving the larger goal of protecting water quality in receiving waters. Additionally, where stormwater characterization monitoring is coupled with targeted stormwater management effectiveness monitoring projects (S8.E), results from monitoring can provide useful information to guide management decisions.

3. Organization and schedule

This section describes the components of the program team and schedule, including special training that will be required and the process of revising this document.

3.1. Roles and responsibilities

Table 1 presents the major aspects of the program and the responsible personnel.

Table 1. Program roles and responsibilities.

Position	Role and responsibility
Department of Ecology Permit Coordinator	Review and approval of Plan and program deliverables from Clark County to Ecology
Program representative for Clark County	Overall management of the County's NPDES Phase I compliance activities. Monitor and assess the quality of work. Comply with corrective action requirements.
Project Manager (PM)	Develop, implement, ensure approval of, and maintain the Plan. Verify the Plan is followed and the program is producing data of known and acceptable quality. Ensure adequate training and supervision of all monitoring and data collection activities. Validate and verify data collected, and initiate corrective action as appropriate. Prepare reports. Manage all project contracts and consultant work. Acquire, verify, and transfer data from field and lab. Manage and oversee monitoring activities, including sampling decisions for targeted storm events, and data management.
Field Staff	Site preparation, equipment set-up, maintenance, sampler set-up and calibration, sample retrieval, sample transfer to laboratory
Laboratory managers	Supervise laboratory personnel involved in generating analytical data for program. Ensure all QA/QC procedures are completed as required and analytical documentation is accurate and complete. Enforce corrective action as required.
Laboratory quality assurance managers	Supervise and verify all aspects of QA/QC in the laboratory. Validate and verify data before released from the laboratory.

3.2. Schedule

Table 2 indicates primarily the original estimated implementation schedule for the stormwater monitoring program, assuming an ongoing program.

Table 2. Original program schedule.

Calendar Year 2009				
Activity	**Anticipated Date of Initiation**	**Anticipated Date of Completion**	**Deliverable**	**Deliverable Due Date**
Program startup	March 2009	Ongoing	Planning; equipment procurement, installation, and testing; staff training	Not reported to Ecology
Continuous hydrologic data	May 2009	Ongoing	Develop baseline rainfall/runoff relationship	Not reported to Ecology
Complete program startup	June 2009	8-16-2009	Monitoring equipment installation and testing	Not reported to Ecology
Dry-season events	8-16-2009	9-30-2009	Stormwater Monitoring Report	3-31-2011
2010 - perpetuity				
Wet-season events	10-1 each year	4-30 each year	Stormwater Monitoring Report	3-31 each year
Sediment sampling	10-1 each year	9-30 each year		
Dry-season events	5-1 each year			
Toxicity sampling (first-flush event)	8-1 each year			
Data validation	As samples are collected	1-30 each year		

3.3. Special training needs

Project staff will require the following training and certification:

- Any field personnel involved with monitoring equipment installation or equipment maintenance requiring confined space entry will complete confined space entry training.
- All field personnel will receive training in sampling equipment operation, maintenance, and calibration procedures.

- All field personnel will receive training in all necessary sample collection, sample handling, and chain of custody procedures for sediment, stormwater grab, and stormwater composite sampling.

In addition to technical training, field personnel will receive guidance that addresses specific monitoring issues that may impact their health and safety. Stormwater sampling conditions are often wet, cold, and poorly lit. Additionally, field crews may be exposed to traffic hazards, biological hazards (e.g. medical waste or fecal matter), disease vectors (e.g. snakes or rats), fall hazards, other hazardous materials, and slippery conditions. All field activities will need to keep safety the top priority.

All laboratories selected for physical and chemical analyses will be fully accredited under the Washington Department of Ecology Laboratory Accreditation Program for the parameters to be analyzed.

3.4. Revisions

Changes to the Plan that are substantial enough to require the Plan to be revised and re-submitted to Ecology for approval are considered external revisions. Smaller changes do not require Ecology approval, and are considered internal revisions. All internal revisions will be documented in a Plan Addendum that will be distributed to all program personnel. If multiple Addendums are prepared, they will be compiled and distributed.

Section B. Type, Quality, and Quantity of Data Needed

This section defines the quality of data required to meet the study goals and objectives, and the sampling process design, which determines the type and quantity of data to be collected.

4. Quality objectives

This section defines the data quality indicators (DQIs) for the monitoring program, as well as the measurement quality objectives that will be utilized to evaluate the quality and usability of stormwater and sediment data. These DQIs will be achieved through attention to the prescribed sampling, measurement, and QA/QC procedures presented in this Plan.

4.1. Data quality indicators

DQIs may be either qualitative or quantitative, and describe the type, quality, and quantity of data that are required to fulfill the program objectives. DQIs utilized to evaluate meeting the objectives for this program include:

- Precision, bias, and accuracy will be evaluated
- Bias will be minimized by using controlled procedures for field sampling, sample handling and processing, laboratory analysis, and record keeping
- Data sensitivity will be upheld by using reporting limits low enough to support stormwater management targets
- Data collected will be of sufficient quality and quantity to enable calculation of event mean concentrations, and seasonal and total annual constituent loads
- Data and samples collected will meet the program-specific requirements for representativeness

4.2. Measurement quality objectives

Measurement quality objectives (MQOs) describe measures of performance and criteria for acceptance, based primarily on DQIs, which provide the basis for evaluating data quality and usability. They are defined separately for hydrologic data and chemical data. They can indicate the minimum threshold levels for measures of bias, precision, accuracy, sensitivity, completeness, and representativeness associated with the data (Table T 6). For chemical data, MQOs often are based upon specific types of quality control (QC) samples that are collected in the field and/or analyzed in the laboratory (Table T 7).

Bias Bias represents a difference from the "true" concentration, and may be either high or low. Potential sources of bias include sampling and analytical procedures that introduce contamination, instability of samples during transportation and storage, interference from other constituents in the sample matrix, inability of the analytical method to measure all forms of the constituent of interest, and faulty calibration of the measurement process. Errors of bias are minimized through use of standardized procedures by properly trained staff. Blank samples and laboratory control samples

provide indications of bias from field and lab procedures. If bias is detected in these QC samples, required actions will be taken as defined in Table T 6.

Precision Precision is a measure of the repeatability of a set of replicated results, and is considered to represent random error in the measurement process. Poor precision is due to difficulties in obtaining samples under identical conditions (e.g. contamination, sub-sampling issues, or variability of field conditions during the time the replicate samples are collected) or poor sensitivity of laboratory procedures. Precision of field samples will be evaluated by either field replicates or splits. Field replicate samples will be collected by near simultaneous grab samples (for fecal coliform and total petroleum hydrocarbons) while splits of flow weighted composite samples will be made under controlled conditions at the analytical laboratory. The two values (sample and replicate or split) will be compared to provide a measure of the relative percent difference between them (RPD). [RPD = $\frac{R_1 - R_2}{[R_1 + R_2]/2}$ $x100$]. It is important to consider the magnitude of replicate error in context of the ambient concentration, e.g. where concentrations are low, the percent difference between replicates may be large even when the absolute magnitude is not. Sampling procedures will be carefully evaluated if field replicate results exceed the defined MQO criteria (Table T6).

Sensitivity Sensitivity is the measure of concentration that is sufficient for the analytical method to positively identify and quantify analytical results. For a given method, the sensitivity is referred to as the detection limit, which can be defined with different levels of precision. The method detection limit is the minimum concentration that can be statistically determined with 99 percent confidence to be greater than zero. Lab reported concentrations less than the method detection limit are reported by the lab as “ND” along with the specific method reporting limit and annotated with a “U”. The method reporting limit is the concentration that the laboratory can report with documented precision and accuracy under routine operating conditions, and is higher than the method detection limit. Concentrations greater than the method detection limit but less than the method reporting limit will be qualified by the laboratory with “J” for estimated. Section 12 contains further discussion of reporting limits.

Accuracy Accuracy is a measure of the agreement between a measured result and the true value of the parameter, i.e. it represents a combination of bias and precision. Laboratory analytical accuracy is measured by analyzing known reference samples or by analyzing samples spiked with known concentrations of the constituents of interest. Factors that influence accuracy include those that cause bias (contamination, sample degradation, matrix interference, laboratory calibration) and those that influence precision (random error in the sampling and analytical process). Accuracy of discharge measurements is related to limitations of the equipment, specifically to the limits of discharge that the sensors are calibrated to operate within. Accuracy of discharge measurements will be assessed by comparing the instantaneous flow or stage measurements at installed controls with the concurrent flow meter readings during several different flow conditions. Maximizing accuracy of discharge is achieved by appropriate selection of measurement technology for the discharge conditions that will occur. Accuracy of tipping bucket rain gages is measured using a burette to meter water into the funnel. Once the bucket tips the volume is recorded. The measurement is repeated 10 times on each side, then averaged and compared to the rain gage specification's tip volume. Specific actions that are required if MQOs are exceeded are defined in Table T6.

Completeness Completeness is defined as the proportion of samples collected relative to the total number planned to be collected. It also depends upon the completion of analytical work by the laboratory. Combined, it represents an assessment of how field and laboratory problems affected the success of the data collection effort. Completeness depends upon the proportion of target storms that are sampled, as well as adequate packing of samples for transport and timely processing at the laboratory.

Data that are qualified but still usable will be counted as valid data for assessing completeness, although data that are rejected for use will not be considered. During the data validation process, an assessment will be made whether enough valid data exist to meet the requirements of the Permit. If not enough valid data are obtained, corrective actions will be initiated by the Project Manager or his/her designee.

Representativeness Representativeness is a measure of the extent to which the sample data accurately and precisely represent site conditions. It is a qualitative measure, although it depends upon meeting quantitative criteria such as the method and timing of the sampling process relative to the storm hydrograph. Additionally, representativeness includes the requirement that samples are collected during a target storm event, and that laboratory analysis be conducted within the method-required holding times. Data will be rejected if the samples are deemed non-representative.

5. Sampling process design

This section describes the sample process design for the monitoring program, including the selection of the locations for monitoring, the approach for targeting storm events, the chemical, physical, and biological parameters to be analyzed, and other measurements to be taken in the field and the laboratory.

Each site will be sampled according to the prescribed frequency unless good faith and good professional practice do not result in collecting a successful sample for the full number of storms. Sixty-seven percent of the forecasted qualifying storms that result in actual qualifying storm events are required to be sampled, up to a maximum of eleven (11) storm events per water year. These storms will be distributed throughout the year to approximately reflect the distribution of rainfall between wet and dry seasons: 60-80% of samples will be collected during the wet season and 20-40% of samples will be collected during the dry season. Additionally, up to a maximum of 3 samples may be analyzed if they are collected in an attempt to sample the eleven required storm events but do not meet the rainfall volume storm event criterion, as long as they do meet the remaining storm event and sample criteria. Accordingly, the maximum number of sampled storm events to be analyzed for stormwater chemistry is fourteen (14) per year. Additionally, a flow-weighted composite sample for toxicity testing will be collected during the first-flush qualifying storm, and an annual composite sediment sample will be collected by in-line sediment trap. A summary of the general sampling design is provided in Table 3.

Table 3. Summary of sampling design.

Sample type	Criteria	Maximum number
Stormwater chemistry Manual grab plus flow-weighted composite	67% of qualifying storms	11
Wet-season samples	60-80% of required samples	7-9
Dry-season samples	20-40% of required samples	2-4
Optional additional samples	Meet all criteria except rainfall volume	3
Total number of stormwater samples	Required plus optional	14
Toxicity	First-flush qualifying storm	1
Sediment chemistry	Composite over sample deployment	1

5.1. Site selection

Locations for stormwater monitoring were selected by evaluation of the following maps, generated by the County's geographic information system (GIS)—

- Stormwater sewer system
- Streets and right-of-way features
- Parcels, land use, and zoning
- Hydrology features and stream classifications
- Geology and soil type
- Topography
- Aerial photography and LIDAR imagery

For all selected sites, 100 percent of the drainage areas and 100 percent of the basin drains to the storm system upstream of the monitoring site were mapped. Field visits were conducted to evaluate the prospective sites regarding the likely basic hydrology and feasibility of monitoring (i.e. access, potential for vandalism, safety issues, and equipment installation requirements). The results of this investigation for the selected monitoring sites are presented in Table 4.

Detailed site visits have been conducted to determine the site-specific requirements for permanent installation and operation of flow monitoring and automatic sampling equipment. Final assessment of the suitability of each site for long-term monitoring will be completed once preliminary hydrologic data have been collected during the implementation phase of the program. Particular attention will be given to evaluating the

characteristics of flow within the pipes, e.g. sufficient depth to collect the sample and maximum velocities that will not cause the sampler to overfill the bottles during intense storm events.

Table 4. Characteristics of selected monitoring sites.

Parameter	**Commercial site**	**High-density residential site**	**Low-density residential site**
Name of basin	GM 34921	MH 5171	Packard tributary channel (LDR010)
Drainage area (acres)	26.77	238.65	43.12
Receiving waterbody	Cougar Creek	Vancouver Lake	Packard Creek
Land use distribution			
High-density residential (%)	17.5	98.9	0
Low-density residential (%)	0	0	100
Agricultural (%)	0	0	0
Parks/Wildlife refuge	0	1.04	0
Commercial (%)	82.5	.04	0
Nearest rain gage	On site	On site	On site
Time of concentration	8 minutes	24 minutes	15 minutes

Time of concentration Time of concentration for each monitoring site was calculated based on methodology described in the Hydraulic Manual published by the Washington State Department of Transportation (Publication M23-03.01, March 2007) using the following equations:

$$T_c = T_{t1} + T_{t2} + ...T_{tn}$$

$$T_t = \frac{L^{1.5}}{K\sqrt{\Delta H}}$$

where: T_C = time of concentration in minutes

T_t = travel time of flow segment in minutes

m = number of segments

L = length of segment in feet

ΔH = elevation change across segment in feet

K = ground cover coefficient in feet

S = slope of segment ($\frac{\Delta H}{L}$) in feet per feet

K values were from the table of Ground Cover Coefficients (Figure 2-5.3) in the Hydraulics Manual. The commercial site includes a single segment comprised of pipe flow for its entire length, and the time of concentration is 8 minutes. The high-density residential site is also a single segment of pipe flow, and the corresponding time of concentration is 24 minutes. The low-density residential site contains two segments (overland flow and roadside ditch), with a total time of concentration of 15 minutes. Further details of the calculations for each site are described in Appendix 1.

5.2. Qualifying storm events

Stormwater samples will be collected and reported based on water year, defined as October 1 through September 30. Two seasons are defined for sampling: wet and dry. Overall qualifying criteria for sampling storms during each season are described in Table 5. Clark County interprets "forecasted qualifying storms" criteria as a minimum of 75% chance of at least 0.25 inches of daily precipitation for Sunday through Friday at Vancouver (using the KVUO long-term weather observation station located at Pearson Airport in Southern Clark County). The latest forecasts of daily total precipitation from 24 hours prior to the day of the anticipated storm(s) are utilized when evaluating "forecasted qualifying storms". Clark County contracts with the Televent Public Safety Edition for commercial online forecasting service. Storms ending on Saturday, that would otherwise be qualifying, typically will not be sampled due to too little time available for sample processing prior to the contracted commercial lab's closure Saturday afternoon through Sunday. However, depending on the timing of the forecasted precipitation, more than a single daily precipitation total is taken into account to address storms that continue into the next day.

Table 5. Criteria for targeting storms for sampling.

Criteria	Wet season	Dry season
Seasonal period	October 1 through April 30	May 1 through September 30
Minimum amount of rainfall	0.20 inches	0.20 inches
Rainfall duration	No fixed minimum	No fixed minimum
Antecedent precipitation conditions	≤0.02 inches in previous 24 hours	≤0.02 inches in previous 72 hours
Inter-event dry period	6 hours	6 hours

The seasonal first-flush storm event is defined as an event in August or September with at least a one-week antecedent dry period (no more than 0.02 inches in previous 168 hours *or* no recorded storm flow at outfall; flow must be measured continuously through period). If a qualifying event does not occur in August or September, or toxicity test results are anomalous or invalid, then a storm can be sampled in October for toxicity testing irrespective of antecedent dry period (S.8.D.2.d).

5.3. Precipitation and discharge monitoring

A continuous rainfall record will be utilized to establish a rainfall/runoff relationship for each monitoring site, as well as to identify the number of qualifying storm events throughout the year. Precipitation data will be acquired from rain gages installed at each sampling location. Alternative nearby rain gage data (Figure 5, site #3, Salmon Creek at NE 156th Street Station) will be used to monitor precipitation in case of malfunction of the site-specific gages. Rain gages will be equipped with telemetry (e.g. land-line telephone, cellular, or radio wiring) to allow for remote data acquisition by modem or internet access. Real-time precipitation data for the alternative precipitation gages can be accessed at the following web link: http://64.4.176.145/ui/SelStationSensor.asp. Field staff will depend on real-time precipitation data to alert them when antecedent and precipitation conditions have been met for a qualifying storm event, so that the sampler may be programmed accordingly. All rain gages will be maintained according to manufacturer's recommendations and guidelines described in Section 8.

Flow rate at each monitoring site will be continuously measured and recorded for the first year, and during each targeted storm event for subsequent years. Flow rate will be

determined using the appropriate flow monitoring technology for the specific site. The key factor to collecting accurate stream gage information is the hydraulic control for the site. In a natural channel (most similar to the low-density residential monitoring site), hydraulic control is the feature that physically affects the water surface elevation. In an engineered pipe system, characteristic of the sites that are monitored inside storm drains, an artificial structure may be required to control the water level. Artificial controls may be built in the channel or selected pipe conveyances for the purpose of simplifying the procedure of obtaining more accurate discharge records. Ensuring the correct design and installation of artificial controls for this study, as appropriate, will be the responsibility of the Project Manager.

5.4. Stormwater sample collection

Two types of stormwater samples will be collected at each monitoring site: (1) manual grab samples, and (2) flow-weighted composite samples. The goal is to sample sixty-seven percent of forecasted qualifying storms, up to a maximum of eleven (11) events per year (S8.D.2). Sampling will be distributed proportionally throughout the year, approximately reflecting the distribution of rainfall, with 60-80% of the samples (i.e. 7-9) collected during the wet season and the remainder 20-40% (i.e. 2-4) collected during the dry season (S8.D.2.a) (Table 6). Additionally, up to three (3) additional samples may be analyzed if they are collected during an attempt to sample the eleven required storm events where the storm does not meet the rainfall volume storm event criterion, as long as other storm event and sample criteria are met. Accordingly, the maximum number of sampled storm events to be analyzed is fourteen (14) per year.

The objective of the stormwater sampling is to sample representative storm runoff. Accordingly, sampling will exclude any base flow that may occur after the end of the storm event. If significant volumes of base flow are observed to occur (i.e. 72 hours after measurable precipitation), periodic base flow samples may be collected if sufficient volume passes the sampler intake. The number and schedule for these samples will be determined by the Project Manager.

Table 6. Approximate yearly distribution of stormwater samples.

Season	Number of samples
Wet (October 1 – April 30)	60-80% (7 – 9)
Dry (May 1 – September 30)	20-40% (2 – 4)
Annual total of required samples	11
Additional optional samples	3
Maximum number of required plus optional samples	14

Grab samples Grab samples are required for analysis of fecal coliform bacteria and total petroleum hydrocarbons (TPH) (S8.D.2.e). Grab samples will be collected manually by, depending on the target analyte, either filling the appropriate containers directly from the outfall at each site or skimming from the water surface. These samples will be collected as early as feasible in the storm event, preferably coincidently with a composite sample. If the logistics of the event prevent field personnel from visiting the sampling sites at the beginning of the storm (i.e. if the storm begins in the middle of the night), grab samples may be taken from different storm events.

Additionally, whenever possible, fecal coliform and TPH will be analyzed from the same grab sample. However, if the necessary limit of holding time for bacterial samples cannot be met (i.e. the laboratory is closed for the weekend or holiday), the grab sample for fecal coliform may be taken later in the storm event. Finally, the grab samples will be collected during storm events that meet all qualifying storm criteria that can be evaluated at the beginning of the event. If the storm subsequently fails to qualify simply because sufficient rainfall does not occur over the course of the event, the grab sample will still be considered valid. The justification is as follows: water quality conditions of the grab sample represent the conditions at the beginning of a storm regardless of the amount of total runoff that occurs.

Composite sampling Composite samples of stormwater will be collected for a range of conventional parameters, nutrient, metals, and organics (Table T 2). These composite samples will be flow-weighted, based on equal sub-sample volumes collected primarily over the targeted storm flow periods. The sampler will be programmed to begin as early as practical during a qualifying storm event, and to continue beyond the estimated time of concentration for the site. The time increments for sub-sampling will be determined

according to the projected time for a defined volume of water to flow past the sampler intake. Each composite sample will contain at least 10 aliquots (samples with 7-9 aliquots will be acceptable only if they meet other criteria and are necessary to achieve a representative balance of seasonal events) (S8.D.2.b). Samples will be collected for a minimum of 75 percent of the storm hydrograph for storms lasting less than 24 hours; if storms last longer than 24 hours, sampling will cover 75 percent of the storm hydrograph during the first 24 hours of the storm.

Sampling may be initiated automatically, based on increasing flow measured by the flow meters, or manually by field personnel based on precipitation data. The automatic function of the sampler will be defined by the ambient level of baseflow (if present) plus the sensitivity of the instrument (approximately 0.03 – 0.1 feet). Composite sampling focuses on sampling representative storm runoff. Accordingly, samples collected after the end of a storm event will be excluded from the composite whenever possible.

If the volume of stormwater sample collected from a qualifying storm is insufficient to allow analysis for all parameters listed in Table T 2, the sample will be analyzed for as many parameters as possible in the following priority order:

A. All land use types:
   i. TSS
   ii. Conductivity
   iii. MBAS
   iv. Metals and hardness

B. Commercial site
   v. PAHs and phthalates
   vi. Pesticides
   vii. Nutrients
   viii. BOD_5
   ix. Chlorides

C. Residential sites
   v. Nutrients
   vi. Pesticides
   vii. PAHs and phthalates

viii. BOD_5

ix. Chlorides

First-flush toxicity sampling A composite sample will be collected for seasonal first-flush toxicity analysis, defined by antecedent conditions described in Section 7.2. This sample will be analyzed for a subset of the full constituent list described for stormwater analysis (marked with * in Table T 2), as follows:

- TSS
- Chloride
- Hardness
- MBAS
- Metals, total dissolved (Cu, Zn, Cd, Pb, and Hg [only at commercial site])
- PAHs
- Phthalates
- Herbicides
- Pesticides

Field personnel will contact the toxicity laboratory prior to the forecasted storm event to inquire about gamete (test organism) availability. If the laboratory confirms that gametes of sufficient quantity and quality are not available for toxicity testing, no attempt will be made to collect toxicity samples for that storm event. A second sample may be collected later in the season, assuming the toxicity event criteria can be met, if the sample cannot be collected as originally defined despite good faith and documented efforts (i.e. because the qualifying storm does not occur) or the toxicity test is invalid or cannot be completed (i.e. because gametes are not available). If the second attempt is also unsuccessful, no further attempt will be made to sample again for that water year; these efforts will be documented in the annual stormwater monitoring report.

The toxicity sample will be collected using the same general autosampler configuration described for composite sampling, and then split into two subsamples: one for toxicity testing and one for chemical analysis. The composite sample may be collected with a time-based rather than a flow-weighted approach if this is necessary in order to collect the required sample volume. Because a minimum of 40 liters is required, it may be

necessary for field crews to switch out and replace bottles during the course of sampling in order to obtain sufficient sample volume.

Sediment sampling Annual sediment samples for physical and chemical analysis (Table T 2) will be collected at each outfall location using an in-line sediment trap. This sediment trap will be constructed with a 1-L amber wide-mouth glass jar that is securely mounted and located in a representative sampling position.

5.5. Monitoring equipment

This section describes the deployment and installation of monitoring equipment: rain gages, flow meters, and automatic sampler. A fully integrated sampler system will be selected instead of multiple components from various manufacturers. The appropriate selection of sampler characteristics for each site is critical to efficient collection of water-quality data. The programming needs of the data collection program have been carefully considered, as well as the use of plug-in modules such as velocity flow modules.

An integrated cell phone telemetered system consisting of a programmable data logger interfacing with ISCO^{TM} Series 6712 sampler(s), site specific flow meter(s), and rain gage will be installed at each monitoring station. Each sampler unit will be configured with custom-built insulated containers sized to hold four (4) 18.5 liter glass bottles (carboys) with appropriate caps and capable of holding adequate ice volume to chill samples.. A separate rain gage nearby will provide input to the system. Flow monitoring will be provided by a suitable sensor integrated with the 6712, depending on the site-specific hydraulic conditions: a bubbler flow module will be installed at those sites that will be intermittently dry and a pressure transducer system will be deployed where water is always present.

All equipment will be installed in a durable instrument shelter located on the surface, which will provide easy access to the samplers during storm events. Each site will be equipped with cell phone telemetry. Telemetry will allow the data to be viewed in real time to facilitate storm tracking. In addition, real-time data will make it possible to efficiently identify erroneous data so that corrective actions can be taken in a timely manner. All equipment in the shelter will be powered with batteries that are continuously charged with local AC power or solar panels during daytime.

Section C. Sampling and measurement procedures

6. Sampling (field) procedures

The quality of field sampling activity is critical to the success of the effort to ensure that sample collection is consistent, the samples adequately represent the target storm conditions, and the data will be comparable to data collected by other monitoring programs. To this end, field operations and sample collection will be carefully planned and implemented, and will follow specific Clark County Standard Operating Procedures (SOPs) that have been or will be developed and SOPs to be developed by other Permittees (Table T 1). This section provides an overview of the field procedures that will be implemented. Specific details are provided in the Clark County SOPs.

All field activities must be documented in detail in field sheets that are assembled into a three-ring binder or field notebooks. Corrections are made by drawing a single line through the error so it remains legible, writing the corrections adjacent to the errors, and initialing the correction so that the data are legally defensible. Notes on the collection and handling of samples should be sufficiently detailed to allow a reviewer's understanding and evaluation of the process.

Required field log entries include the following—

- Name of program, and location of field work
- Identity of field crew
- Sequence of events
- Any changes to the sampling plan
- Site and climatic conditions
- Date, time, location, identification, and description of each sample collected
- Instrument calibration procedures
- Field measurement results
- Description of QC samples collected, if appropriate
- Unusual circumstances that may affect interpretation of data, if appropriate

6.1. Precipitation and flow monitoring

The schedule for maintenance of precipitation and discharge monitoring equipment may vary depending on the specific instrumentation at each site, but will follow the general pattern described in Table 7.

Table 7. Precipitation and flow monitoring maintenance schedule.

Monitoring station	Timeframe	Frequency	Activities
Precipitation	First year of operation	At least monthly and during all site visits	Inspect and service equipment
	After first year	Every other month, once telemetry is functioning	Inspect and service equipment
Discharge	First year of operation	At least monthly	Inspect and check sensor
	After first year	Every other month, once telemetry is functioning	Inspect and service sensor

Data from the rain gages and flow meters will be automatically downloaded daily just after midnight and as needed during and following sampled events. Rain gages and flow sensors will be inspected monthly and serviced as needed during the first year of operation, and every other month after the first year. Calibration of stage sensors will be conducted at least monthly to establish accurate rainfall/runoff relationships. Stage calibration will be conducted less frequently, as appropriate, in subsequent years.

6.2. Sample collection procedures

This section describes the procedures for identifying a storm for sampling, setting up the monitoring equipment prior to the storm, collecting the manual grab sample, monitoring sample collection and retrieving the sample, validating the sample collected, and maintaining equipment between storm events.

Each sample site will have a checklist of the samples to be collected, the required equipment and supplies, instructions for retrieving the bottles and processing the samples, and detailed servicing instructions for setting and programming the sampler for the next series of sample collection.

Storm targeting During the period when potential storms are suspected to qualify for sampling, the Project Manager (or designee) will consult precipitation forecasts for Clark

County to identify storms during the next 3-7 day period that may meet the minimum target criteria. If forecasts indicate that a target storm may be approaching, the Project Manager (or designee) will decide whether to consider sampling. Once a decision is made to sample the event, as appropriate, County staff will continue reviewing the forecasts daily. Internet forecasts will be archived to document the decision process.

Ideally, the Project Manager (or designee) will make the decision about targeting the storm for sampling no less than twenty-four hours prior to the onset of the storm. If the storm is targeted for first-flush toxicity, the Project Manager will immediately contact the Toxicity Laboratory to confirm that materials are available for the toxicity test.

The Project Manager, or designee, will be responsible for assembling the field team to conduct pre-storm site setup activities and managing all field and sampling decisions until the samples are delivered to the laboratories.

Pre-storm site setup If feasible, within 24 hours of the onset of the target storm, the field team will visit each site to prepare for sampling. Based on the forecasted precipitation quantity, storm duration, and estimated runoff volume, the Project Manager or designee will have determined the appropriate sampler pacing rate. During the setup visit, the field team will conduct any necessary maintenance, place the clean sample bottles into the autosampler, and program the sampler. All setup activities will be recorded on a field sheet. The following tasks will be performed—

- Inspect status of precipitation gage and stage / flow sensor and clean as appropriate
- Check status of sampler pump tubing, rinse with lab-grade deionized water and calibrate sampler pump as appropriate
- Stock sampler with clean glass bottles and ice
- Confirm sampler and flow meter programs and settings
- Run sampler diagnostics to ensure correct operation of distributor arm
- Start sampler program and ensure program is active and disabled, waiting for signal from flow meter to begin collecting the sample
- Ensure batteries are charged

Grab sample collection The Project Manager or designee will keep close watch on the precipitation gages to confirm that the targeted storm event meets all qualifying criteria,

keeping in close contact with the field team to keep them informed. The grab sample should be collected as early as feasible with the onset of storm runoff. All activities and observations should be recorded on the same field sheet used to document pre-storm setup activities.

The following tasks should be conducted at each site—

- Check status of autosampler and battery; troubleshoot as needed
- Collect grab samples by dipping sample containers into flow stream of outfall (skim samples from the surface for TPH)
- Collect replicate and/or blank grab samples as scheduled
- Label and store samples on ice for transport to laboratories

Mid-event site visit (as appropriate) During the course of the storm event, the Project Manager or designee will closely track the cumulative precipitation and discharge in order to ascertain if sampler bottles will fill before the end of the storm. This judgment will be based upon the expected runoff from observed rainfall, the sampler pacing rate, and the amount of precipitation that is forecast. If it is determined that the sampler bottles may fill too soon, the Project Manager or designee will send the field crew out to check the sampler and exchange sample bottles if needed. The following activities will be conducted—

- Check the status of the autosampler and decide if sampler bottles should be exchanged or if additional ice is needed
- If bottles are exchanged—
    - Review sampler report and record data
    - Cap and label bottles and store on ice for transport and storage
    - Deploy new clean bottles into the sampler
    - Restart the sampler program
- Check batteries and replace if necessary

All activities and observations should be recorded on the same field sheet used to document pre-storm setup activities.

Composite sample retrieval Throughout the storm event, the Project Manager or designee will monitor weather conditions and forecasts to identify when the event has

ended. Once the storm is over, the field team will be mobilized as soon as feasible to retrieve the composite sample. The following tasks will be conducted—

- Review the sampler report and record data
- Cap and label bottles and store on ice for transport
- Collect equipment blank sample, if scheduled
- Power down the autosampler or place in stand-by mode

All activities and observations should be recorded on the same field sheet used to document pre-storm setup activities.

Composite sample splitting and processing In order to minimize potential contamination and holding times, all sample splitting and processing will be done at a nearby contracted accredited commercial analytical lab. As soon as feasible after sampling is completed, the bottles of flow-weighted composite samples will be quickly placed within ice-filled coolers, to minimize sample's exposure to light, and transported to the laboratory for splitting, processing, and analyses. Composite sample splitting and processing procedures are described in detail in the USGS National Field Manual for the Collection of Water-Quality Data (NFM, 2002, Chapter 5). Sample-specific requirements for volume, containers, and hold times are described in Table T3. This section provides a brief overview of sample bottles as well as sample splitting and processing procedures.

Glass composite bottles (18.5 liter) for samples to be analyzed for inorganic, organic and metal constituents should be ordered pre-cleaned, acid rinsed, and should arrive capped from the laboratory. Before deploying in the field, all bottles (including acid-rinsed bottles) should be pre-rinsed three times with deionized water and stored in plastic coolers.

At the accredited laboratory, samples will be divided through a Teflon^{TM} cone splitter (Figure 6) or, depending on volume, just a Teflon^{TM} churn splitter (for set-up instructions, see NFM, Chapter 2). Composite samples exceeding 14 liters are first split using a Teflon cone splitter in order to get a homogenous and representative smaller sample that is then split using a 14 liter Teflon churn splitter into required laboratory sample analyses containers. All tubes exiting the cone splitter must be the same length, as short as possible, and pre-cleaned. Organic-compound samples require Teflon^{TM} tubing. The

entire composite sample volume must be poured or pumped into the splitter from the large automatic sampler bottles for thorough distribution, taking care not to spill any sample or overfill analytical sample bottles.

All field sample bottles must be clearly labeled with the following information—

- Station number
- Date and time

Bottle labels may be written directly on each bottle with permanent, waterproof marker or preprinted labels that will remain securely attached to the bottles, even if they become wet.

Document on field forms pertinent sampling information including the types of samples collected, last sample time, number of aliquots, estimated total sample volume, and sampler screen readings.

Equipment cleaning Cleaning of all sample splitting and processing equipment will be done at the contracted accredited analytical lab. Dissemble and clean the cone splitter before reusing (NFM, Chapter 3). If the cone splitter will not be reused immediately, clean and rinse with deionized water then place in a plastic bag.

Equipment should be cleaned in an area protected from airborne or other sources of contamination. All equipment should be inspected prior to cleaning for stains, cuts, or abrasions. Any parts that are not in good working order should be replaced. Equipment should first be rinsed with deionized water then cleaned as follows—

- Wear gloves that are chemically resistant to any cleaning solvent being used
- Detergent wash and rinse with tap water
  - 0.1 to 0.2-percent solution
  - Nonphosphate laboratory-grade detergent
- Acid rinse
  - 5-percent hydrochloric acid in wash bottle
  - Rinse exterior of equipment
  - Pump acid solution through tubing
  - Do not reuse acid solution
- Rinse with deionized water

- Methanol rinse
  - Pesticide-grade methanol
  - Dispense from a Teflon™ wash bottle or pump through tubing
  - Discharge used solvent into appropriate waste container for flammable liquids

Equipment blank and field blank data should be examined regularly to determine whether adjustments to the cleaning protocol are needed.

Transport of samples to lab and chain of custody Samples should be packaged and transported to the laboratory for analysis as soon as possible, preferably on the day of collection (assuming that the laboratory will be open to receive the samples on the appropriate day). Samples designated for chilling will be packed with ice in insulated ice chests. The amount of ice will depend upon the length of time in transit and the ambient air temperature.

Before transporting samples, check that sample bottles are labeled correctly and that a chain-of-custody form has been completed. Samples should be packed carefully in the ice cooler to prevent bottle breakage, container leakage, and sample degradation. Check that bottle caps have been securely fastened.

In order to provide continuous documentation of each time the sample changes hands, a chain-of-custody form will be completed by the person sending the samples. These forms must describe the location, date, sample number, nature of sampling program, and name of individuals who collected the samples, as well as their signature. Chain-of-custody forms will follow the samples through the analytical process in the laboratory, with personnel sign-offs each time the samples change hands.

6.3. Field sample validation

The Project Manager or designee will validate the samples using the qualifying storm criteria, as well as the following—

- Grab sample
  - Check that necessary antecedent precipitation conditions existed
  - Review field forms and the storm hydrograph to confirm the grab sample was collected as early as feasible in the storm

  - Review field forms for any anomalous conditions that occurred that would disqualify the grab sample
  - Check the grab sample containers to ensure they are filled correctly
- Composite sample
  - Determine that the storm met the criteria for a qualifying event
  - Review field forms for missed aliquots or bottles overfilling or other conditions that would disqualify the composite sample
  - Confirm that the composite sample contains at least 10 aliquots
  - Check that sufficient volume was collected to complete laboratory analysis
  - Review storm hydrograph and timing of sample aliquot collection to confirm that the sample represents at least 75 percent of the hydrograph during the first 24 hours of the storm

6.4. Preventative maintenance

Periodically, preventative maintenance will be conducted between storms so that equipment can be properly operated. All maintenance issues, including any sign of vandalism, will be comprehensively documented in field notebooks associated with each site. Approximately every six months, the sampler suction line will be replaced for cleaning. Typically before each sampler set-up, the sampler will be manually checked and calibrated as needed to verify that the sampler is pumping the desired volume and that the sampler is working correctly.

6.5. Sediment sample collection

Suspended sediment samples will be collected with an in-line sediment trap that is continuously deployed for a target of about one year during each annual monitoring period. Field personnel will deploy the sediment trap at the end of the outfall pipe for each site, as described in Section 7.4. Crews will regularly inspect the sediment sample bottle over the deployment period to make note of the volume of material that is accumulating in the trap, and to ensure that the container opening is not collecting litter or debris. The frequency of inspections will be increased as appropriate, depending on the speed of accumulation of sediment in the trap.

When the sediment trap is nearly full, or at the end of the planned deployment period, the collection bottle will be carefully removed from the housing and capped with a screw lid.

The bottle will be placed on ice for transport to the analytical laboratory for further processing. In order to preserve the grain size distribution, the sediment sample should never be frozen prior to analysis.

7. Measurement (laboratory) procedures

This section describes the analytical and toxicological laboratories that will conduct analyses for this program, as well as the analytical methods that will be used.

7.1. Analytical laboratories for chemical analysis of stormwater and sediment

Columbia Analytical Services (CAS) laboratories are currently certified to perform environmental analysis of stormwater by the Washington State Department of Ecology through the Environmental Laboratory Accreditation Program (ELAP) (#C1203 / C544, expires July 8, 2011). Most water and sediment samples will be analyzed at Columbia Analytical Services main laboratory in Kelso, Washington but the firm also has a network of different laboratory facilities located throughout the United States. The CAS laboratory located in nearby Kelso is one of the largest CAS laboratory facilities, offering a wide variety of environmental analyses and analytical methods. A full list of parameters that CAS laboratory is accredited to perform is found at

http://www.ecy.wa.gov/apps/eap/acclabs/lab.asp?id=1203

The CAS Kelso laboratory is located at the following address:

Columbia Analytical Services
1317 S. 13th Ave
Kelso, WA 98626
360-577-7222
www.caslab.com

Analyses for some fungicides and herbicides (Pentachlorophenol, Triclopyr, 2,4-D and MCPP) will be conducted by Edge Analytical Laboratories, which is currently accredited by Ecology (#C567-11, expires January 18, 2012). A full list of parameters that Edge Analytical Laboratories is accredited to perform is found at

http://www.ecy.wa.gov/apps/eap/acclabs/lab.asp?id=1251

Edge Analytical Laboratories applicable lab is located at the following address:

Edge Analytical Laboratories, Inc
1620 S Walnut St
Burlington, WA 98233
1-800-755-9295
http://www.edgeanalytical.com/

7.2. Toxicological Laboratory

Toxicological tests will be performed by Nautilus Environmental, which is currently certified (#C587, expires February 8, 2011) to conduct bioassay tests , including freshwater chronic toxicity tests, by the Washington State Department of Ecology. A complete list of parameters that Nautilus Environmental is accredited to perform is found at http://www.ecy.wa.gov/apps/eap/acclabs/lab.asp?id=1294

Nautilus Environmental is located at the following address in Washington:

5009 Pacific Highway East, Suite 2-0
Tacoma, WA 98424
http://www.nautilusenvironmental.com/

7.3. Chemical analysis and toxicity testing procedures

This section describes the analytical methods and toxicity testing procedures that will be utilized for analysis of stormwater, sediment, and toxicity samples collected as part of this program.

Chemical analysis of water and sediment All analytical instruments used for analyses for this program will be maintained and calibrated regularly according to specifications in the internal laboratory Quality Assurance Plan (QAP), including all appropriate standard operating procedures (SOPs), instrument manufacturer's specifications, and specific method requirements.

Target constituents and corresponding analytical or Ecology approved alternative methods, method detection limits, method reporting limits, and reporting limit targets for stormwater and sediments are described in Table T 4 and T5. Alternative methods have been reviewed and approved by Ecology.

Toxicity testing procedures This section describes the freshwater chronic toxicity test that will be performed for seasonal first-flush stormwater samples. Specifications for

invalid or anomalous tests are defined, as well as follow-up actions when toxicity is detected, and yolk testing to augment chemical analysis of the sample matrix.

Analytical instruments

All analytical instruments used for analyses for this program will be maintained and calibrated regularly according to specifications in the internal laboratory Quality Assurance Plan (QAP), including all appropriate standard operating procedures (SOPs), instrument manufacturer's specifications, and specific method requirements.

Toxicity testing method

Testing of the annual first-flush composite sample for toxicity must begin within 36 hours of sample collection. The test method is based on Environment Canada Biological Test Method (Toxicity tests using early life stages of salmonids species, rainbow trout, EPS 1/RM/28, 2nd edition, July 1998), with modifications by Canaria et el., 1998 (Canaria, Elphick, and Bailey, 1999. A simplified procedure for conducting small-scale short-term embryo toxicity tests with salmonids. Env. Toxicol, 14: 301-307).

After the end of the 7-day exposure period, the surviving eggs will be placed into glass vials for visual examination for viability. Eggs failing to complete morphogenesis are considered non-viable. Several endpoints are defined for this test—

- Death of eggs during exposure
- Unsuccessful fertilization
- Inhibited development of eggs

An EC_{50} will be calculated for each result using the Spearman-Karber Method; Abbot's correction may be applied. At least five concentrations plus a control sample will be evaluated. If an EC_{50} result equals 100% sample or less, the permit requires follow-up actions (described below).

Invalid or anomalous tests

If the laboratory does not follow the test protocol or the results do not meet acceptability criteria as defined in the protocol, the toxicity test results are considered invalid. The

laboratory is responsible for identifying invalid test results and arranging to repeat them. Invalid tests may also be identified by Ecology.

Anomalous results occur if the toxicity test is conducted according to the protocol but the results are identified as suspect by the anomalous test identification criteria defined by Ecology (Publication # WQ-R-95-80, *Laboratory guidance and whole effluent toxicity test review criteria*). Test results may only be identified as anomalous by Ecology.

All invalid or anomalous tests will be repeated with freshly sampled stormwater. Invalid or anomalous results will be reported in the annual stormwater monitoring report.

Follow-up actions if toxicity is observed

If toxicity is detected, the following actions will be taken—

- Terminated organisms will be preserved for 6 months
- Chemical analysis results will be compared to fish embryo toxicity test results to identify toxicants detected within 60 days of the sample date
- The presence of a detected parameter within the range identified as toxic will be determined (reference source literature)

Follow-up actions in response to toxicity detection may also include adding other chemicals to the analysis list, if adding them might help identify the toxicants that are present. Additions to the list of analytes should be based upon knowledge of industrial or commercial activity that suggests the presence of the specific chemical, which has been previously identified as toxic.

All results and follow-up actions must be documented and submitted to Ecology in the annual stormwater monitoring report.

Additional testing procedures

Chemical analysis of yolks from the highest test concentration may be conducted if toxicity is detected but concurrent chemical analysis of the sample does not detect any chemicals at sufficient concentration to be toxic to fish embryos. Gas chromatograph/mass spectrometer (GC/MS) analysis of the embryo or yolk must only be sufficient to qualitatively identify contaminants present, which may then be added to the standard constituent list of analytes for quantitative analysis.

The GS/MS analysis should expand beyond the list of standard chemicals monitored by the Phase I Permit in an effort to identify as many as reasonably possible. The detection of a chemical within the yolk or embryo implies significant exposure that should be evaluated in the literature to see if it has been identified as toxic.

Section D. Quality Assurance and Quality Control Procedures

8. Quality Control

This section describes the quality control (QC) requirements for all field and laboratory activities conducted by this program. Data quality will be evaluated according to the measurement quality objectives (MQOs) described in the Quality Objectives Section.

8.1. Field quality control

Field QC requirements include procedures for field measurement and documentation, sample collection, field QC samples, and corrective action for identified issues for field activities.

Field quality control procedures All preventative maintenance and calibration, as appropriate, of field and sampling equipment, including rain gages, flow sensors, automatic samplers, and other field instrumentation will be conducted according to the schedule defined in the Sampling (Field) Procedures Section.

Designated binders will be maintained for original field records for all monitoring and field activities. Program-specific forms will include storm data field sheets and maintenance inspection/activity sheets. Documentation will also include details surrounding sediment trap deployment, work permits for confined spaces, and chain-of-custody forms. All entries in field documentation must be clearly written ; errors will be corrected by crossing a line through the error and entering the correct information (dated and initialed).

All sampling and processing procedures will be conducted according to the specifications detailed in this Plan, in order to ensure consistent and representative samples and reduce contamination.

Usually, field QC samples will be collected during storm events as described below, with the exact schedule to be determined by the Project Manager. QC samples must be clearly identified and labeled with unique sample identification codes so that the results can be clearly tracked. QC sample data will provide the basis for necessary corrective action to be taken to correct problems that may occur, as determined by the Project Manager. The process of corrective action will include the identification of a problem, action to correct the problem, monitoring to evaluate the corrective action, verification that the problem has been appropriately solved, and documentation of the chain of events. Corrective actions may deal with issues in chain-of-custody, sampling collection and/or transport to the laboratory, field record keeping, or additional training that may be required. Field procedures may be revised as appropriate to deal with problems that are identified.

Field quality control samples Field QC samples are used to evaluate sample collection procedures and other conditions that could potentially impact the integrity of the samples before they reach the laboratory Table T 8. These could include ambient site conditions during sample collection, the processes of storage and transport, and the adequacy of equipment cleaning. The target ratio of QC samples to ambient samples will be at least 1:20, distributed across S8D and S8F sampling sites. Specific types of QC samples include—

- Field Split samples
- Field Replicate samples
- Field Blank samples
    - Transport blanks
    - Transfer blanks
    - Equipment (rinsate) blanks
    - Temperature blanks

Field Splits Split samples provide a measure of the variability of the analytical process. Split samples will be prepared at the laboratory by reserving a second carboy of sample water immediately following processing the large initial volume through the cone splitter.

The second carboy will be processed through the churn splitter and analyzed in the same manner as the sample. Up to 10 split samples will be analyzed per year as follows: at least two splits (to include the first split collected each year) will be analyzed for the entire suite of chemical constituents. Based on available sample volume, up to eight additional splits will be run for only one group of analytes selected from among: conventionals, nutrients, metals, and organics. Results from split samples will not be utilized to accept or reject data, but as a qualitative assessment of sampling precision.

Field Replicates Field replicates provide a measure of the total variability of the sampling and analytical process. One or two replicate samples per year will be collected concurrently with early storm grab samples and analyzed for fecal coliform and total petroleum hydrocarbons. Field replicates will be collected for grab samples by filling a second set of bottles in rapid succession to the first.

Results from replicate samples will not be utilized to accept or reject data, but as a qualitative assessment of sampling precision. If the relative percent difference between replicate samples exceeds 25 percent (except 50 percent for fecal coliform samples and other samples with ambient concentrations near the detection limit, which are inherently more variable), the field crew will be notified to identify and correct the source of sampling variability as appropriate.

Transport blanks Transport blanks provide a measure of possible contamination from a variety of sources associated with transporting samples. Prior to sample collection on a particular date, blank water that has been certified for the suite of constituents to be analyzed will be used to fill a randomly selected set of sample containers in the office prior to leaving for the field and submitted to the lab for analysis. Transport blank samples will be clearly identified by a unique sample identification code associated with a specific site. A single transport blank will be prepared once per year.

Transfer blanks Transfer blanks are prepared in the field by transferring certified blank water to a laboratory- supplied sample bottle. The transfer blank detects contamination due to handling, the designated sampling area, or contamination present in a laboratory-

supplied bottle. A transfer blank will be collected at least once per year at a representative field site.

Field equipment (rinsate) blanks Field equipment blanks are collected to specifically identify the potential for contamination of the sampling equipment and evaluate the effectiveness of cleaning procedures. An equipment blank will be collected at least once per year from an automatic sampler within a representative configuration of equipment used by this program. The equipment blank sample will be collected in-situ by pumping the appropriate certified blank water through the cleaned sampler into a randomly selected set of bottles, labeling them as appropriate, and submitting the bottles to the laboratory for analysis. Equipment blank results will be evaluated in the same manner as field blanks, and similar corresponding actions will be taken as appropriate (Table T6).

Temperature blanks Temperature blanks provide a measure of the temperature conditions during transport to the laboratory, and thereby the potential for sample degradation prior to analysis. Temperature blanks will be prepared with deionized water and submitted in the coolers for all environmental samples delivered to the laboratory for this program. The laboratory will be responsible to check the temperature of the temperature blanks upon receipt. If the temperature is greater then $6^{o}C$, the field crew will be notified so that appropriate corrective actions (i.e. additional ice packed in the coolers) can be taken prior to the next sample.

Field blanks (including transport, transfer, field equipment rinsate, and temperature blanks) will be evaluated and appropriate corrective actions taken as needed (Table T6). If the results for field blank samples exceed twice the method reporting limit for any constituent, the data for those analytes will be flagged as appropriate based on the ambient concentrations observed (Table T4 and T6). Additional corrective action will include notification of the field crew and evaluation of sampling procedures in order to identify the source of the contamination. Additionally, if laboratory blank results indicate that the significant source is from the field, not the lab, field blanks may be collected for subsequent samples to help identify and minimize possible sources of contamination.

8.2. Laboratory quality control

The analytical laboratories will conduct all chemical and physical analyses as requested, meeting holding time requirements and target reporting limits to the extent possible.

Columbia Analytical Services analyses are performed according to the laboratory's NELAP – approved quality assurance program. Test results meet requirements of the current NELAP standards (under the NELAC Institute) except when noted otherwise in individual laboratory result case narratives.

Data acquired from laboratory QC samples will be utilized to evaluate the quality of analytical data, to determine the need for appropriate corrective action, and to interpret the effectiveness of corrective actions after they are implemented. Each method SOP includes a QC section that describes the minimum QC requirements for that procedure. Internal QC checks may differ but all procedures include both negative and positive controls (e.g. blank samples or standard reference samples, respectively), as described in the following sections.

Method blanks Method blanks are defined as certified blank water that is carried through the entire sample preparation and analysis procedure. All reagents are added in the exact manner as in analyzing environmental samples. At least one method blank is required for each batch of samples for each method. The purpose of the method blank is to evaluate the potential for contamination during preparation, processing, and analysis for each batch.

The method blank is processed along with, and under the same conditions as the associated samples including all steps of the analytical procedure. The goal is no detectable contaminants. If contamination is observed, the source must be investigated and measures taken to eliminate or minimize the problem. Samples that are associated with a contaminated method blank are evaluated to determine the best corrective action (i.e. reprocessing or flagging with appropriate qualifying codes).

Laboratory control samples Laboratory control samples (LCS) include standard reference samples, reagent spikes, surrogate spikes, and continuing calibration verification standards. The LCS are utilized to assess the performance of the total analytical system, including all preparation and analysis steps. The number of LCS

samples varies for each method and is specified by the SOP. Analyte concentrations in the LCS must be within the calibration range of the methods where possible.

Data from these samples are compared to established criteria by the use of control charts, and indicate that the analytical system is out of specification if they exceed these criteria. Criteria for determining the need for corrective action are based upon guidance from the National Environmental Laboratory Accreditation Conference (http://www.epa.gov/nelac/standard/2003nelacstandard.pdf). Data from any samples associated with LCS results that exceed these criteria are reanalyzed or qualified as appropriate.

Reagent spike samples consist of water that is known to be free of analytes of interest, and subsequently spiked with known and verified concentrations of target constituents. Generally, because of concern for analyte cross-reactions and interferences, reagent spike samples are made from a subset of the entire method list. This subset is selected to represent the method performance characteristics for all the analytes measured by the method.

Surrogates are frequently used in organic chromatography analyses and are selected to represent the variety of chemistries for the selected analytes of the method. They are added prior to sample preparation and extraction, and provide a measure of recovery for each sample matrix. They are deliberately chosen because they are unlikely to be present already as an environmental contaminant in the sample. Frequently surrogates are compounds that are isotopically labeled, fluorinated, or brominated.

Generally, the number of surrogates varies with each method. When used, surrogate compounds are added to all samples in the set, including standards and blanks.

Certified reference materials are commercially purchased and contain verified concentrations of analytes, used to measure analyte recoveries for applicable methods.

Continuing calibration verification standards are used to check instrument response to the analytes in relation to the calibration curve. They are prepared similarly to calibration standards at a concentration typically in the midrange of the calibration curve.

8.3. Toxicity testing quality control

Toxicity tests will meet quality assurance criteria as defined in the Environment Canada Manual EPS 1/RM/28 and the Department of Ecology publication # WQ-R-95-80 (Laboratory guidance and whole effluent toxicity test review criteria).

In general, the toxicity test shall consist of five concentrations plus one control sample, with four replicates for each concentration. Additionally, seven replicates of 100% samples shall be evaluated in order to provide material for yolk and embryo analysis if necessary. Test concentrations shall be determined by using a dilution factor of 0.5.

9. Data Management Procedures

Data from this monitoring program will consist of the following—

- field activity data, including sampling and equipment maintenance documentation
- field monitoring data, including flow, precipitation, and field probe data
- laboratory data, including water quality, toxicity, and sediment data

All records of monitoring information must be retained for at least five (5) years, including all calibration and maintenance records, all original recordings for continuous monitoring instrumentation, copies of all required reports, and records of all data.

9.1. Field activity data

Field notebooks and 3-ring binders will contain all field activity data, as follows—

- completed storm data field sheets
- chain of custody forms
- maintenance inspection and activity logs

Field staff is responsible for updating this information and storing it in appropriate binders securely located and available to the Project Manager.

9.2. Field monitoring data

Field data will be downloaded remotely every day automatically via cell phone telemetry onto a project computer in backed up data files. The files are then appended and converted for storage, analyses, and editing utilizing commercially available ISCO Flowlink or Aquaticinformatics Aquarius software, and an industry-standard Microsoft

Access database. Rainfall data and discharge data can also be downloaded on demand remotely, as needed, and imported into a project database. The Project Manager is responsible for acquiring the data in a timely manner and for maintaining the project database, which will be backed up regularly.

9.3. Laboratory data

All laboratory data will be transmitted electronically from the laboratory to the County's Project Manager or designee. The Project Manager is responsible for acquiring the data and conducting preliminary quality checks. The Project Manager also is responsible for conducting final review, verification, and validation of data.

Section E. Assessment Procedures

10. Audits and Reports

This section describes the processes that will ensure that the quality assurance procedures specified by this Plan are being implemented correctly, that the quality of the data is acceptable, and that corrective actions are conducted in a timely manner.

10.1. Audits

Audits are an important tool to verify that the quality assurance procedures described in this Plan are being adequately implemented as necessary. During an audit, the reviewer will check for the following—

- Sufficient documentation of all required activity
- Compliance with site-specific plans
- Correction of any problems that have been identified

Audits may be scheduled by request from the Project Manager at Clark County or the Department of Ecology.

Laboratory audits The Department of Ecology's Laboratory Accreditation Unit conducts external audits of the Columbia Analytical Services Kelso laboratory facility to verify that all laboratory operations comply with established requirements. The audits are conducted with about a month's notice every 2-3 years. The audit is implemented by trained and qualified personnel who are independent of the audited activity. The audits

verify that laboratory procedures follow published methods, approved SOPs, and the overall quality management requirements of the laboratory. A major function of these audits is to ensure that established policies are documented in unit SOPs, are well understood by personnel, and are fully integrated into the workflow. Any nonconformances to documented procedures that are identified are reported in the audit report and necessitate corrective action by appropriate personnel.

Field audits The Project Manager or a designated representative may conduct audits of the field activities, if deemed appropriate. This audit will address the following questions—

- Are field operations performed as specified?
- Are documentation records complete?
- Are problems identified and resolved appropriately?
- Are data management procedures adequate?

Any nonconformance to established protocols will result in appropriate corrective action. The Project Manager has ultimate responsibility for ensuring that the corrective action response is appropriate, complete, and documented.

10.2. Reports

Reports that will be generated for this program include storm files, status reports, and the Annual Stormwater Monitoring Report.

Storm files Storm files are not required by the Phase I Permit, but will be maintained to track the details of each stormwater sampling event. They will include all supporting information and documentation of the event, including the rainfall and runoff flow rate, the stormwater hydrograph showing when samples were collected, and documentation of weather tracking and forecasts. Other supporting documentation may include notes regarding sampling anomalies or other issues that may impact data analysis and reporting. The Project Manager, in collaboration with the field staff, is responsible for maintaining these files.

Status reports Status reports will be prepared and submitted to the Program Representative for Clark County as requested. These will include a summary of the

successful and valid samples to date, a summary of the quality control results for analytical data, and a discussion of any issues that may need to be addressed. The Project Manager is responsible for preparing these reports.

Annual stormwater monitoring report The Stormwater Monitoring Report is required to be submitted with the annual report every year (S8.H). Each report will contain all monitoring data collected during the previous water year (October 1 through September 30). As appropriate, the reports will integrate data from earlier years into the analysis of results.

Reports will be submitted in both paper and electronic form and will include:

- A summary description of each monitoring site, describing location, land use, drainage area, and hydrology
- A comprehensive data report, including QA/QC, with an explanation and discussion of the results
- The annual and seasonal (wet and dry) pollutant loads for each constituent at each site, expressed in total pounds and pounds/acre, based on all completely monitored seasons

The stormwater monitoring summary description will include any new basin information that is not presented in this Plan, such as basin changes that may affect the hydrology or pollutant loading at the site.

The data report will explain the program results, present the data, document the overall quality of the data (i.e. completeness, representativeness, and usability), and discuss any data anomalies. The QA/QC section will include a summary and discussion of the field and laboratory QC procedures and results, and whether the measurement quality objectives were met. This section will include any planned changes to the current Plan that may be appropriate to address QA issues that have been observed.

Pollutant loads will be calculated for both wet and dry seasons; the annual (water year) load will be determined by the sum of the seasonal loads. Pollutant loads will be determined from runoff volume based on continuous flow monitoring and the flow-weighted composite analytical data.

11. Data verification and validation

This section describes the data review, verification and validation procedures that determine whether the data conform to the criteria required by the program objectives.

11.1. Summary of procedures

Data review is the process of examining the data for errors or omissions. Data verification is based on the QC results, and determines whether the data meet acceptance criteria. Data validation includes the complete sampling process to assess whether the appropriate procedures were followed in sample collection.

All data generated by this program will be reviewed and verified for conformance to the requirements of the program. Data will then be validated according to the data quality objectives described in the Quality Objectives Section. Once data are found to be supported by acceptable QC data and to meet the specified measurement quality objectives, they will be considered acceptable and usable for the program.

Procedures for verification and validation will generally be conducted according to the guidance provided by EPA, 2002 (*Guidance on Environmental Data Verification and Data Validation*, EPA QA/G-8). The Project Manager, in collaboration with the support staff, is responsible for ensuring that field data are reviewed and verified. After each successful sampling event, the Project Manager or designee will review rainfall and flow data for gross error (e.g. spikes or data gaps) to verify the completeness of the data. The field staff will also check to see that stormwater samples were collected in accordance with required criteria.

The personnel and management of the analytical laboratories are responsible for the validation and verification of the data generated by each lab. The Project Manager is responsible for ensuring that all analytical data are reviewed once they are transmitted from the laboratory prior to being imported into the County's water quality database. The Project Manager is responsible for verifying that all data meet the QA criteria, and also for validating that all data conform to the objectives of the program and are suitable for reporting.

11.2. Methods of verification and validation

This section presents a brief overview of the methods that may be used for verifying and validating data, including the input that will be necessary, the specific methods to be used, and the output from the verification process.

Data input A variety of records will be necessary for data verification and validation. These could include, but are not limited to, the following—

- Field logs
- Chain-of-custody forms
- Laboratory temperature logs
- Analytical service request
- QC results

Data verification methods Data verification methods will be documented throughout the course of the process, and may be revised as appropriate to the situation. These methods will include—

- Identification of data gaps or missing data
- Identification of data anomalies or spikes outside the limits of reality
- Cross checks with field sheets and calibration records
- Evaluation of expected patterns versus observed
- Evaluation of QC results (within acceptable criteria)

Data verification products Outputs from the data verification process include both the verified data and the verification documentation. Verified data will be submitted to the final quality or usability assessment described in the next section. Verification documentation may be included in status reports and/or the QA/QC section of the Annual Stormwater Monitoring Report. Data verification documentation should also include a discussion of any technical issues or other shortcomings of the data.

12. Data Quality Assessment

Once data have been verified and validated, the final data quality assessment is conducted. The data quality objectives defined in this Plan must be satisfied in order for the data to be considered usable for meeting program objectives. The main goals of this

assessment are to determine if the data meet the quantity of samples required, are representative of stormwater runoff conditions in the basin and the target storm events, and are sufficient to calculate seasonal pollutant loads.

12.1. Data usability assessment

Usability assessment is a qualitative process where data are evaluated relative to the specified measurement quality objectives. Three levels of data quality will be defined—

- Accepted: data meet all requirements for sample collection and analysis, quality control criteria, and documentation
- Qualified: data meet most requirements with only minor deviations from specified methods, critical QC criteria and documentation are met
- Rejected: data do not meet critical QC criteria and documentation is missing

The usability assessment includes examination of outliers, verification that the data are complete, representative, and comparable to other data collected in similar programs.

Data processing guidelines Data processing guidelines will be developed, depending upon the specific data issues that occur. All laboratory results less than the method detection limit (MDL) will be reported as ND and qualified with a "U" flag. For data analyses purposes, ND results will be replaced with one-half of the method reporting limit (MRL) value. Results that are greater than the MDL but less than the MRL will be qualified with a "J" flag.

12.2. Data reduction methods

Data will be used to calculated event mean concentrations (EMCs), seasonal pollutant loads for wet and dry seasons, and total annual pollutant loads. The EMC is the flow-weighted concentration averaged over the duration of a single runoff event. For composite samples that are collected in proportion to flow rate, as specified for this program, the concentration of the single sample for each event represents the EMC. Seasonal and annual loads will be expressed as pounds per year, and pounds per acre per year.

Where base flow occurs, the total annual load is the sum of base flow and storm flow loads. At present, the occurrence of base flow is not well understood at these sites. The

following steps for dealing with base flow will be utilized if significant base flow is observed.

Determination of base flows and volumes Base flow will be determined from the record of continuous flow measurements. Flow occurring after 72 hours with no measurable rainfall ($\leq$ 0.02 inches) may be indicative of base flow. Integration of base flow over the wet season and dry season will provide seasonal base flow volume. The sum of seasonal base flows will provide the total annual base flow.

Determination of flow-weighted mean base flow concentrations Chemical concentrations of sampled base flow events during the monitoring year will be tabulated, and a flow-weighted mean base flow concentration will be calculated. If seasonal differences in base flow quality are observed, mean concentrations will be calculated separately for the wet and dry season. Otherwise, a flow-weighted mean concentration will be calculated for the entire monitoring year.

Determination of storm flows and volumes The total combined flow (including base flow, if present, and storm flow) will be determined from the continuous flow measurements. Total flow will be integrated over the time period of interest and corrected with base flows previously determined, if appropriate, to determine seasonal and total annual storm flow.

Separation of storm flow and base flow in sampled storm events For each sampled storm event where base flow has been observed, the base flow contribution will be subtracted from the total flow to determine the storm flow contribution. These components of the sampled storm event will be normalized to the fractional volumetric contributions so that sum of each fraction is equal to one.

Calculation of storm flow EMC Where base flow does not occur, the storm flow event mean concentration (EMC) is simply the measured concentration derived from laboratory analysis of a flow-weighted composite sample. Where base flow occurs, the total storm event concentration contains a mixture of storm and base flow components. The base flow component will be separated from the total concentration using the principles of mass balance in order to calculate the EMC for the storm component alone. This

calculation utilizes the volumetric fractions previously obtained and the mean base flow concentration determined above:

$$EMC_{sf} = \left[EMC_{tot} - (C_b * f_b)\right] / f_s$$

where EMC_{sf} = EMC for storm flow component

EMC_{tot} = EMC derived from laboratory analysis

C_b = mean base flow concentration

$f_{b \text{ or } s}$ = volumetric fractions for base flow and storm flow, respectively

Determination of flow-weighted mean storm flow concentration A flow-weighted mean storm flow concentration will be determined from storm flow EMCs. If seasonal differences in stormwater quality are observed, mean storm flow concentrations will be calculated separately for wet and dry seasons. Otherwise, the flow-weighted mean storm flow concentration will be determined for the entire monitoring year.

Estimation of storm flow loads Mass loads from storm flow will be estimated by the product of total storm flow volumes and annual flow-weighted mean storm flow concentrations.

13. Data analysis and presentation

This section describes the content of the Annual Stormwater Monitoring Report that presents the stormwater characterization monitoring component and includes data collected during the previous water year. The following four elements are required to be included—

- Site summary
- Data and QA/QC report
- Annual pollutant loads, based on water year, for each site (total pounds and pounds/acre)
- Seasonal pollutant loads for each site, based on water year, for each site (total pounds and pounds/acre)

13.1. Site summary

Each site summary will describe the location, land use, drainage basin area, and hydrology. Additionally, the report will document all land use changes in the drainage basin for each site that could potentially impact either the hydrology or pollutant loading.

13.2. Data summary

This section will include the following elements—

- Stormwater sampling
    - Documentation of whether sampled storms met the target criteria
    - Summary of the number of qualifying storms that were sampled, additional events sampled that did not meet all criteria, ratio of events sampled during each season
    - Time series graphs for each sampled storm of precipitation, flow rate, and initiation of aliquot collection for each sample
    - Table of analytical results for each sampled storm
    - Table of annual and seasonal loads for each constituent
- Toxicity testing
    - Documentation that sampled event met the qualifying criteria
    - Bench sheets for toxicity test results
    - Analytical reports for toxicity and chemistry data
    - A follow-up report, as appropriate, including summary of literature comparison related to toxicity test results
    - Documentation, as appropriate, for all invalid and anomalous test results
- Sediment sample
    - Table of sediment data
- Program changes
    - Overview of any changes proposed to be made to the sampling program
    - List of constituents with concentrations consistently less than detection limits after two years of analysis (which will be dropped from the program)

13.3. QA/QC summary

This section will include the following elements—

- Data validation information for each sample that describes any field and/or laboratory QC issues and corrective actions, as appropriate

- Summary QA report, including
    - Overview narrative summarizing all data validation information
    - General assessment of usability and representativeness of data
    - Description of any proposed changes to the existing QAPP to deal with observed QA/QC problems

Table T 1. Standard Operating Procedures list

Category	Title	Author
Sampler Setup and Gauge Calibration / Maintenance	Sampler Setup Procedures for S8D Land use / Stormwater Characterization Sites	Clark County Clean Water Program
	Sampler Setup Procedures for S8F BMP / Stormwater Treatment Effectiveness Sites	Clark County Clean Water Program
	Pressure Sensor Calibration SOP Memorandum	Herrera Environmental Consultants, Inc
	Continuous Gauge Maintenance SOP Memorandum	Herrera Environmental Consultants, Inc
Storm Analyses and Sampler Pacing	S8 Storm Analyses and Sampler Pacing Procedures	Clark County Clean Water Program
	S8 Wetland#1 and Wetland#2 Independent Inflow and Outflow Seasonal Random Sample Dates	Clark County Clean Water Program
Data Retrieval and Data Logger Programs	Data Retrieval SOP Memorandum	Herrera Environmental Consultants, Inc
	Clark County NPDES Monitoring Data Logger Program Description	Herrera Environmental Consultants, Inc
Sediment Sampling	SOP for Collection of Stormwater Sediments Using In-Line Sediment Traps	City of Tacoma and Washington State Department of Ecology

Table T 2. Parameters analyzed in stormwater and sediment.

Parameters analyzed in stormwater from flow-weighted composite samples [* parameters analyzed in sample collected for first-flush toxicity testing]
Conventionals
Total suspended solids (TSS) *
Turbidity
Conductivity
Chloride *
BOD_5
Hardness *
Methylene blue activating substances (MBAS) *
Nutrients
Total phosphorus
Orthophosphate
Total Kjeldahl nitrogen
Nitrite-nitrate
Metals
Total recoverable (Cu, Zn, Pb, Cd, Hg) *
Dissolved (Cu, Zn, Pb, Cd, Hg) *
Organics
Polycyclic aromatic hydrocarbons (PAHs) *
Phthalates *
Herbicides (2,4-D, MCPP, Triclopyr) *, Fungicides, Pentachlorophenol *
Insecticides, nitrogen (Prometon*), Other Insecticides, Dichlobenil*
Insecticides, organo-P (Diazinon, Malathion, Chlorpyrifos) *
Parameters analyzed in stormwater from grab samples
Bacteria
Fecal coliform
Total Petroleum Hydrocarbons (TPH)
Diesel range organics (NWTPH-Dx)
Gasoline range organics (NWTPH-Gx)
Parameters analyzed in sediment
Conventionals
Total solids
Total organic carbon
Grain size
Metals
Total recoverable (Cu, Zn, Pb, Cd, Hg)
Organics
Polycyclic aromatic hydrocarbons (PAHs)
Phthalates
Phenolics
Polychlorinated biphenyls (PCBs)
Fungicides (Pentachlorophenol)
Insecticides, organo-P (Diazinon, Malathion, Chlorpyrifos)

Table T 3. Sample information for chemical analyses for stormwater.

Parameters analyzed in stormwater	Volume (mL)	Glass Composite / Grab Sample Container	Holding time
Conventionals			
TSS	1000	Glass composite carboy, acid rinsed	7 days
Turbidity	50	Glass composite carboy, acid rinsed	48 hours
Conductivity	20	Glass composite carboy, acid rinsed	28 days
Chloride	5	Glass composite carboy, acid rinsed	28 days
BOD_5	500	Glass composite carboy, acid rinsed	48 hours
Hardness	Calc.	Glass composite carboy, acid rinsed	14 days
MBAS	1,000	Glass composite carboy, acid rinsed	48 hours
Nutrients			
Total phosphorus	100	Glass composite carboy, acid rinsed	28 days
Orthophosphate	5	Glass composite carboy, acid rinsed	48 hours
Total Kjeldahl nitrogen	25	Glass composite carboy, acid rinsed	28 days
Nitrite-nitrate	5	Glass composite carboy, acid rinsed	28 days
Metals			
Total recoverable (Cu, Zn, Pb, Cd)	500	Glass composite carboy, acid rinsed	6 months
Dissolved (Cu, Zn, Pb, Cd)	500	Glass composite carboy, acid rinsed	6 months
Total recoverable (Hg)	500	Glass composite carboy, acid rinsed	28 days
Dissolved (Hg)	500	Glass composite carboy, acid rinsed	24 hrs to filter & preserve / 28 days
Organics			
PAHs	1,000	Glass composite carboy, acid rinsed	7 days for extraction, 40 days for analysis
Phthalates	1,000	Glass composite carboy, acid rinsed	
Herbicides and Fungicides	1,000	Glass composite carboy, acid rinsed	
Insecticides, nitrogen	1,000	Glass composite carboy, acid rinsed	
Other Insecticides including, organo-P	2,000	Glass composite carboy, acid rinsed	
Bacteria			
Fecal coliform	100	Grab - Sealed sanitary polystyrene	24 hours
Total Petroleum Hydrocarbons			
Diesel (NWTPH-Dx0	500	Baked amber glass	7 days until extraction, 40 days after extraction
Gasoline (NWTPH-Gx)	120	glass	14 days
Toxicity	45-60	Glass composite carboy, acid rinsed	24 hours

Table T 4. Information on analytical methods and reporting limits for target constituents in stormwater.

Parameter	Laboratory Method	Method Detection Limit	Method Reporting Limit	Target Reporting Limit
Conventionals				
Total suspended solids (TSS)	SM 2540D	1 mg/L	1 to 10 mg/L (depends on degree that sample clogs filter)	1 mg/L
Turbidity	EPA 180.1	0.04 NTU	0.20 NTU	0.2 NTU
Conductivity	SM 2510 B	0.4 umho/cm	2 umho/cm	1 umho/cm
Chloride	EPA 300.0	0.008 mg/L	0.2 mg/L	0.2 mg/L
BOD_5	SM 5210 B	1.0 mg/L	4 mg/L	2 mg/L
Hardness, as $CaCO_3$	SM 2340C	0.03 mg/L	2.0 mg/L	1 mg/L
Methylene blue activating substances (MBAS)	SM 5540C	0.05 mg/L	0.05 mg/L	0.025 mg/L
Nutrients				
Total phosphorus	EPA 365.3	0.004 mg/L	0.01 mg/L	0.01 mg/L
Orthophosphate	EPA 365.3	0.004 mg/l	0.01 mg/L	0.01 mg/L
Total Kjeldahl nitrogen	ASTM D 1426-93B	0.08 mg/L	0.2 mg/L	0.5 mg/L
Nitrite-nitrate	EPA 353.2	0.009 mg/L	0.05 mg/L	0.01 mg/L
Bacteria				
Fecal coliform	SM 9221 E	NA	2 MPN / 100 mL	2 colonies / 100mL
Metals				
Total recoverable copper	EPA 200.8	0.02 ug/L	0.1 ug/L	0.1 ug/L
Dissolved copper		0.02 ug/L	0.1 ug/L	0.1 ug/L
Total recoverable zinc		0.2 ug/L	0.5 ug/L	5 ug/L
Dissolved zinc		0.2 ug/L	0.5 ug/L	1 ug/L
Total recoverable lead		0.005 ug/L	0.02 ug/L	0.1 ug/L
Dissolved lead		0.005 ug/L	0.02 ug/L	0.1 ug/L
Total recoverable cadmium		0.005 ug/L	0.02 ug/L	0.1 ug/L
Dissolved cadmium		0.005 ug/L	0.02 ug/L	0.01 ug/L
Total recoverable mercury	EPA 7470A	0.02 ug/L	0.2 ug/L	0.1 ug/L
Dissolved mercury		0.02 ug/L	0.2 ug/L	0.1 ug/L
Organics				
Polycyclic aromatic hydrocarbons (PAHs)	EPA 8270D			0.1 ug/L
Naphthalene		0.022 ug/L	0.2 ug/L	
2-Methylnaphthalene		0.026 ug/L	0.2 ug/L	

Parameter	Laboratory Method	Method Detection Limit	Method Reporting Limit	Target Reporting Limit
Acenaphthylene		0.015 ug/L	0.2 ug/L	
Dibenzofuran		0.018 ug/L	0.2 ug/L	
Fluorene		0.027 ug/L	0.2 ug/L	
Phenanthrene		0.022 ug/L	0.2 ug/L	
Anthracene		0.024 ug/L	0.2 ug/L	
Fluoranthene		0.02 ug/L	0.2 ug/L	
Pyrene		0.019 ug/L	0.2 ug/L	
Benzo(a)anthracene		0.018 ug/L	0.2 ug/L	
Chrysene		0.028 ug/L	0.2 ug/L	
Benzo(b)fluoranthene		0.017 ug/L	0.2 ug/L	
Benzo(k)fluoranthene		0.024 ug/L	0.2 ug/L	
Benzo(a)pyrene		0.031 ug/L	0.2 ug/L	
Indeno(1,2,3-cd)pyrene		0.021 ug/L	0.2 ug/L	
Dibenzo(a,h)anthracene		0.017 ug/L	0.2 ug/L	
Benzo(g,h,i)perylene		0.019 ug/L	0.2 ug/L	
Acenaphthene		0.026 ug/L	0.2 ug/L	
Phthalates	EPA 8270D			1 ug/L
Dimethyl phthalate		0.021 ug/L	0.2 ug/L	
Diethyl phthalate		0.012 ug/L	0.2 ug/L	
Di-n-butyl phthalate		0.023 ug/L	0.2 ug/L	
Butyl benzyl phthalate		0.018 ug/L	0.2 ug/L	
Bis(2-ethylhexyl) phthalate		0.13 ug/L	1.0 ug/L	
Di-n-octyl phthalate		0.018 ug/L	0.2 ug/L	
Herbicides	EPA 8151A			0.01-1 ug/L
2,4-D		0.05 ug/L	0.4 ug/L	
MCPP		0.02 ug/L	0.1 ug/L	
Triclopyr		0.02 ug/L	0.1 ug/L	
Fungacides	EPA 8151A			
Pentacholorophenol		0.02 ug/L	0.1 ug/L	0.01-1 ug/L
Insecticides	EPA 525.2			
Dichlobenil		0.007 ug/L	0.1 ug/L	0.01-1 ug/L
Insecticides, nitrogen	EPA 525.2			0.01-1 ug/L
Prometon		0.007 ug /L	0.10 ug/L	
Insecticides, organo-P	EPA 8141A			0.01-1 ug/L
Diazinon		0.025 ug/L	0.3 ug/L	
Malathion		0.017 ug/L	0.3 ug/L	
Chlorpyrifos		0.036 ug/L	0.3 ug/L	
Total Petroleum Hydrocarbons				
Diesel	NWTPH-Dx	12 ug/L	250 ug/L	250-500ug/L
Gasoline	NWTPH-Gx	13 ug/L	250 ug/L	250 ug/L

Table T 5. Information on analytical methods and reporting limits for target constituents in sediment.

Parameter	Laboratory Method	Method Detection Limit	Method Reporting Limit	Target Reporting Limit
Total solids	SM 2540B			NA
Total organic carbon	EPA 9060	0.02 %	0.05 %	0.10 %
Grain size	ASTM D 422M			NA
Total recoverable copper	EPA 200.8	0.08mg//Kg	0.1 mg/Kg	0.1 mg/Kg
Total recoverable zinc		0.2 mg/Kg	0.5 mg/Kg	5.0 mg/Kg
Total recoverable lead		0.009mg/Kg	0.05mg/Kg	0.1 mg/Kg
Total recoverable cadmium		0.004mg/Kg	0.02mg/Kg	0.1 mg/Kg
Total recoverable mercury	EPA 7471A	0.002 mg/Kg	0.02 mg/kg	0.1 mg/Kg
Polycyclic aromatic hydrocarbons (PAHs)	EPA 8270D	1.3 ug/Kg	10-20 ug/Kg	70 ug/Kg
Naphthalene		2.3 ug/Kg	10.0 ug/Kg	
2-Methylnaphthalene		2.2 ug/Kg	10.0 ug/Kg	
Acenaphthylene		1.2 ug/Kg	10.0 ug/Kg	
Dibenzofuran		1.2 ug/Kg	10.0 ug/Kg	
Fluorene		1.1 ug/Kg	10.0 ug/Kg	
Phenanthrene		1.4 ug/Kg	10.0 ug/Kg	
Anthracene		1.6 ug/Kg	10.0 ug/Kg	
Fluoranthene		1.6 ug/Kg	10.0 ug/Kg	
Pyrene		1.5 ug/Kg	10.0 ug/Kg	
Benzo(a)anthracene		1.7 ug/Kg	10.0 ug/Kg	
Chrysene		1.5 ug/Kg	10.0 ug/Kg	
Benzo(b)fluoranthene		1.2 ug/Kg	10.0 ug/Kg	
Benzo(k)fluoranthene		1.4 ug/Kg	10.0 ug/Kg	
Benzo(a)pyrene		1.7 ug/Kg	10.0 ug/Kg	
Indeno(1,2,3-cd)pyrene		1.5 ug/Kg	10.0 ug/Kg	
Dibenzo(a,h)anthracene		1.5 ug/Kg	10.0 ug/Kg	
Benzo(g,h,i)perylene		1.5 ug/Kg	10.0 ug/Kg	
Acenaphthene		1.4 ug/Kg	10.0 ug/Kg	
Phthalates	EPA 8270D			70 ug/Kg
Dimethyl phthalate		1.0 ug/Kg	10 ug/Kg	
Diethyl phthalate		1.3 ug/Kg	10 ug/Kg	
Di-n-butyl phthalate		7.9 ug/Kg	20 ug/Kg	
Butyl benzyl phthalate		3.2 ug/Kg	10 ug/Kg	
Bis(2-ethylhexyl) phthalate		7.0 ug/Kg	100 ug/Kg	
Di-n-octyl phthalate		1.7 ug/Kg	10 ug/Kg	

Parameter	Laboratory Method	Method Detection Limit	Method Reporting Limit	Target Reporting Limit
Phenolics	EPA 8270D	2.0 ug/Kg	30 ug/Kg	70 ug/Kg
Phenol		2.0 ug/Kg	30 ug/Kg	
2-Chlorophenol		2.0 ug/Kg	10 ug/Kg	
2-Methylphenol		1.5 ug/Kg	10 ug/Kg	
4-Methylphenol		1.5 ug/Kg	10 ug/Kg	
2-Nitrophenol		1.5 ug/Kg	10 ug/Kg	
2,4-Dimethylphenol		5.5 ug/Kg	50 ug/Kg	
2,4-Dichlorophenol		1.0 ug/Kg	10 ug/Kg	
4-Chloro-3-methylphenol		1.4 ug/Kg	10 ug/Kg	
2,4,6-Trichlorophenol		1.4 ug/Kg	10 ug/Kg	
2,4-Dinitrophenol		17 ug/Kg	200 ug/Kg	
4-Nitrophenol		18 ug/Kg	100 ug/Kg	
PCBs	EPA 8082			80 ug/Kg
Aroclor 1016		1.3 ug/Kg	10 ug/Kg	
Aroclor 1221		1.3 ug/Kg	20 ug/Kg	
Aroclor1232		1.3 ug/Kg	10 ug/Kg	
Aroclor1242		1.3 ug/Kg	10 ug/Kg	
Aroclor1248		1.3 ug/Kg	10 ug/Kg	
Aroclor1254		1.3 ug/Kg	10 ug/Kg	
Aroclor 1260		1.3 ug/Kg	10 ug/Kg	
Fungicides	EPA 8151			1 ug/Kg
Pentachlorophenol		20 ug/Kg	100 ug/Kg	
Insecticides, organo-P	EPA 8141A			
Diazinon		7.4 ug/Kg	50 ug/Kg	50 ug/Kg
Malathion		0.28 ug/Kg	50 ug/Kg	25 ug/Kg
Chlorpyrifos		0.62 ug/Kg	50 ug/Kg	25 ug/Kg

Table T 6. Criteria for measurement quality objectives (MQO) and required actions when not met.

Data Quality Indicator	Evaluation	MQO (criterion)	Action
Bias	Field Blank samples: Transport, Transfer, Equipment (Rinsate)	< 2x method reporting limit (Table T 4 and T 5)	If MQO not met, flag sample result U for any sample < 5x field blank value. Notify field crew for evaluation of sampling procedures.
	Laboratory control samples ,Matrix spikes, Surrogates	% recovery[1] Table T 7	If MQO not met, flag associated sample results J
Precision	Field replicate samples	RPD[2] Table T 7	If MQO not met, flag associated sample results J
	Field split samples	RPD[2] Table T 7	
	Laboratory replicate samples	RPD[2] Table T 7	
Accuracy	Precipitation depth	Within 5% tipping bucket volume	Utilize data from nearest alternative precipitation gage
	Discharge	Within 20% of actual flow for flows in 10%-90% of equip. operating range; within 35% for flows outside this range	Flag event J if Q does not meet MQO
Sensitivity	Reporting limits	Table T 4 and T5	Results < MDL reported as ND and qualified with a "U" flag; Results > MDL but < MRL qualified with a "J" flag
Complete-ness	Chemical data	90% of intended samples	Criterion not met if successfully-collected data < MQO based on amount intended, document performance and significant issues
	Hydrologic data	90% of data record	
Representa-tiveness	Methodology	Analytical and field methods defined	Deviations must be approved by Project Manager
	Holding time (stormwater only)	Table T3	Flag sample J if MQO exceeded by < 2x holding time; Flag sample R if MQO exceeded by 2x or greater
	Storm event criteria	Table 5	Discard sample if storm does not meet MQO
	Sampler performance	At least 10 aliquots, collected for at least 75% of storm hydrograph (or 75% of first 24 hrs of storm hydrograph)	Discard sample if performance does not meet MQO. Deviations must be approved by Project Manager.

[1] Percent recovery = $\frac{SSR - SR}{SA} x100$ where SSR=spiked sample result, SR=sample result, SA=spiked concentration added to spiked sample.

[2] Relative Percent Difference (RPD) = $(\frac{R_1 - R_2}{[R_1 + R_2]/2} \; x100)$

Table T 7. MQOs for analytical bias, precision, and accuracy.

	Bias			Precision	
Quality Control Sample Type (Limits Used)	**Lab control sample (Percent recovery limits)**	**Matrix spikes (Percent recovery limits)**	**Surrogate standards (Percent recovery limits)**	**Lab replicate samples or duplicate lab control samples (Relative percent difference)**	**Field replicate and split samples (Relative percent difference)**
Parameter					
Conventionals					
TSS	80-115	NA	NA	20%	25%
Turbidity	82-109				
Conductivity	88-110				
Chloride	80-120	80-120			
BOD_5	85-115	NA			
Hardness	90-116				
MBAS	65-130	60-130			
Nutrients					
Total phosphorus	85-115	50-144	NA	20%	25%
Orthophosphate	89-118	81-119			
Total Kjeldahl nitrogen	75-125	37-158			
Nitrite-nitrate	88-110	86-117			
Bacteria					
Fecal coliform		NA	NA	NA	50%
Metals					
Total recoverable (Cu, Zn, Pb, Cd, Hg)	85 -115	70 -130 (Hg 76-126)	NA	20%	25%
Dissolved (Cu, Zn, Pb, Cd, Hg)					
Organics					
Pesticides, nitrogen					25%
Dichlobenil	70-130	70-130	70-130	30%	
Prometon	70-130	70-130	70-130		
Pesticides, organo-P					
Diazinon	46-164		Varies with surrogate (43-148)	30%	
Chloropyrifos	42-158				
Malathion	34-146				
Triclopyr	70-130	65-135		35%	

2,4,-D	60-120	60-120	61-129		
MCPP	48-126	48-126			
Pentachlorophenol	69-123	69-123			
Phthalates					
Dimethyl Phthalate	46-111	37-113	Varies with surrogate (26-127)	30%	25%
Diethyl Phthalate	47-113	37-115			
Di-n-butyl Phthalate	44-126	34-119			
Butyl Benzyl Phthalate	48-115	35-119			
Bis(2-ethylhexyl) Phthalate	45-122	33-122			
Di-n-octyl Phthalate	48-119	33-123			
PAHs					
Naphthalene	31-94	16-108	Varies with surrogate (26-127)	30%	25%
2-Methylnaphthalene	29-92	17-100			
Acenaphthylene	36-107	21-117			
Acenaphthene	36-101	22-115			
Dibenzofuran	37-103	26-110			
Fluorene	39-106	26-115			
Phenanthrene	45-104	30-113			
Anthracene	41-103	18-116			
Fluoranthene	46-109	25-120			
Pyrene	46-108	29-114			
Benz (a) anthracene	47-105	24-119			
Chrysene	49-105	27-120			
Benzo(b)fluoranthene	48-108	26-123			
Benzo(k)fluoranthene	49-107	25-122			
Benzo(a)pyrene	42-109	17-123			
Indeno(1,2,3-cd)pyrene	47-111	21-129			
Dibenz(a,h)anthracene	47-110	24-125			
Benzo(g,h,i)perylene	47-109	23-123			
Diesel and Gasoline					
NWTPH-Dx	46-140		50-150	30%	25%
Residual Range Organics	45-159		50-150		
NWTPH-Gx	80-119		50-150		

Table T 8 Field Quality Control Samples

Field QC Sample Type	Frequency and Method
Field Replicate	One or two replicate samples per year will be collected concurrently with early storm grab samples and analyzed for fecal coliform and total petroleum hydrocarbons. Field replicates will be collected for grab samples by filling a second set of bottles in rapid succession to the first.
Field Splits	Split samples will be prepared at the laboratory by reserving a second carboy of sample water immediately following processing through the cone splitter. The second carboy will be processed through the churn splitter and analyzed in the same manner as the original carboy. Up to 10 split samples will be analyzed per year as follows: two splits (to include the first split collected each year) will be analyzed for the entire suite of chemical constituents. Up to eight additional splits will be run for only one group of analytes selected from among: conventionals, nutrients, metals, and organics.
Transport Blanks	One transport blank will be collected each year at a representative site by filling a random set of sample containers (labeled for a specific site) in the office with lab deionized water then carried unopened to the field and back on a sample retrieval run and submitted with them to the lab for analysis.
Field Equipment (rinsate) Blanks	An equipment (rinsate) blank will be collected at least once per year from the automatic sampler within a representative configuration of equipment used by this program. The equipment blank sample will be collected in-situ by pumping lab deionized water through the cleaned sampler into a randomly selected set of bottles, labeling them as appropriate, and submitting the bottles to the laboratory for analyses.
Transfer Blanks	A transfer blank will be collected at least once per year at a representative field site. Transfer blanks are prepared in the field by transferring lab deionized water to a laboratory-supplied sample bottle, labeling them as appropriate, and submitting the bottle to the laboratory for analyses.
Temperature Blanks	Temperature blanks, provided by the contracted lab, will be submitted with all samples to the lab to check that temperature conditions within coolers are properly maintained during transport.

Figure 1. Clark County Washington, with outfall sites for designated land use monitoring basins.

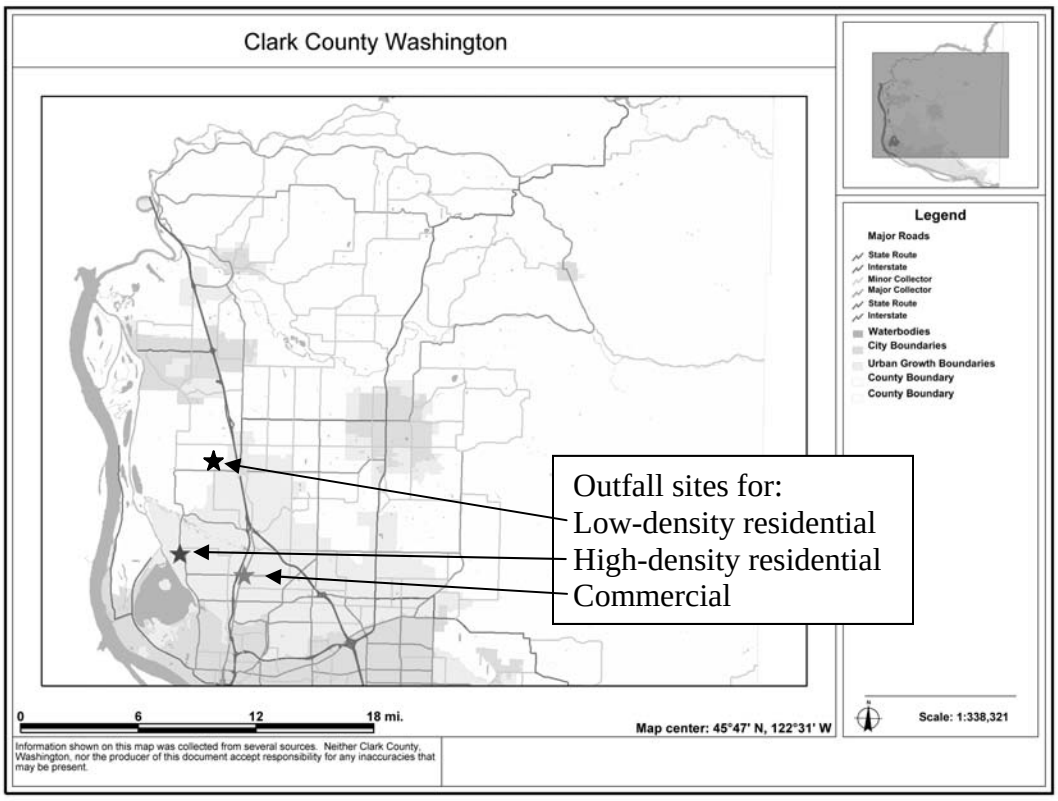


Figure 2. Orthophotography of monitoring drainage basin for commercial site.



Figure 3. Orthophotography of stormwater monitoring drainage basin for high-density residential site.



Figure 4. Orthophotography of stormwater monitoring drainage basin for low-density residential site.



Figure 5. Locations of alternative precipitation gages maintained by Clark County.



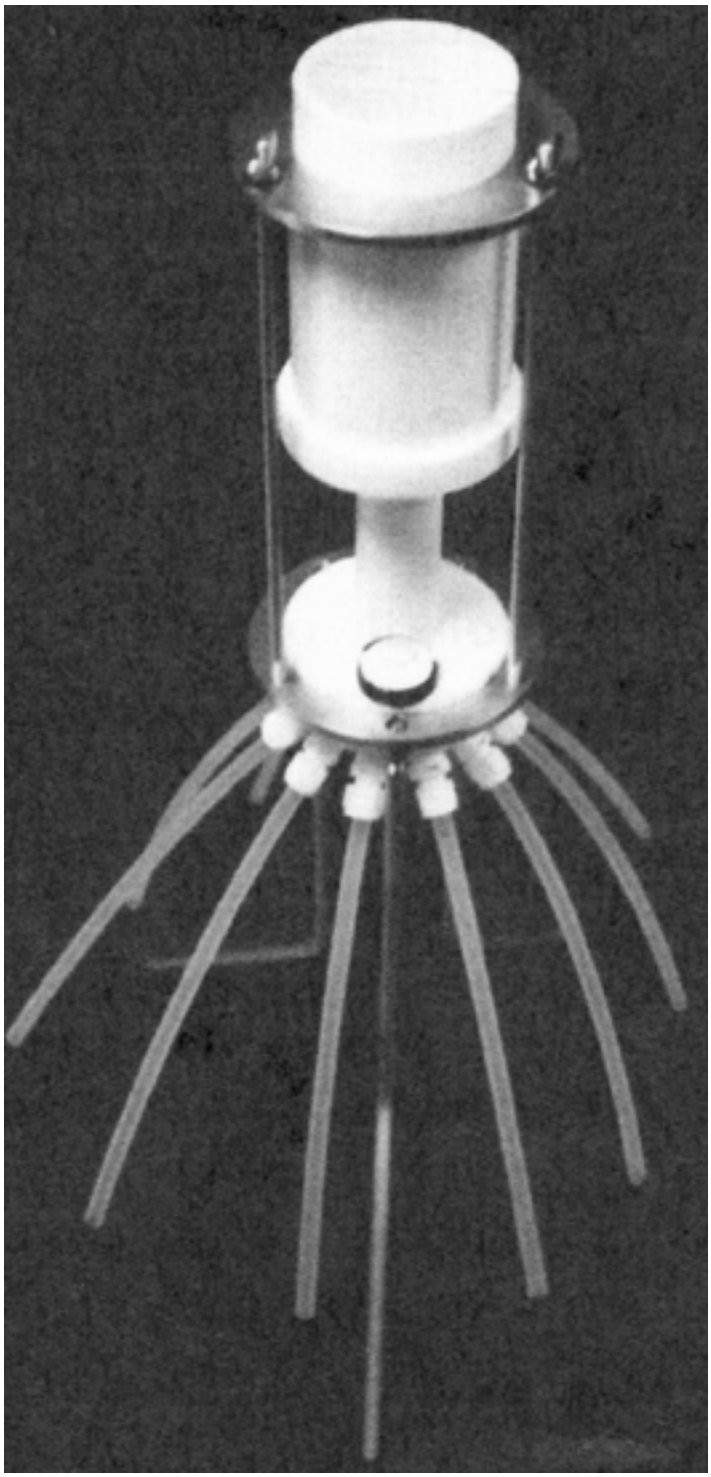


Figure 6. Cone splitter.

Appendix 1 Time of Concentration Calculations

$$T_c = T_{t1} + T_{t2} + ...T_{tn}$$

$$T_t = \frac{L^{1.5}}{K\sqrt{\Delta H}}$$

where: T_C = time of concentration in minutes

T_t = travel time of flow segment in minutes

m = number of segments

L = length of segment in feet

ΔH = elevation change across segment in feet

K = ground cover coefficient in feet

S = slope of segment ($\frac{\Delta H}{L}$) in feet per feet

A. Commercial site
   1. Pipe flow segment for entire length
      a. m = 1
      b. L = 2,970 feet
      c. ΔH = 1.2%
      d. K = 3,450 (average of 3,000 and 3,900)
      e. T_C = 8 minutes

B. High-density residential site
   1. Pipe flow segment for entire length
      a. m = 1
      b. L = 5,300 feet
      c. ΔH = 4.2 %
      d. K = 3,450
      e. T_C = 24 minutes

C. Low-density residential site
   1. Overland flow for first segment
      a. m = 2
      b. L = 700 feet
      c. ΔH = 4%
      d. K = 420
   2. Roadside ditch for second segment
      a. L = 1,270 feet
      b. ΔH = 4%
      c. K = 900
      d. T_C = 8 + 7 minutes = 15 minutes